MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, RANDOM ACCESS, DIGITAL, KEY CODE BYF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic random access memory module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 8.3 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BYF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/80-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for V_{OUT} (low loaded) apply only for a power supply voltage of 7 V dc +0, -1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

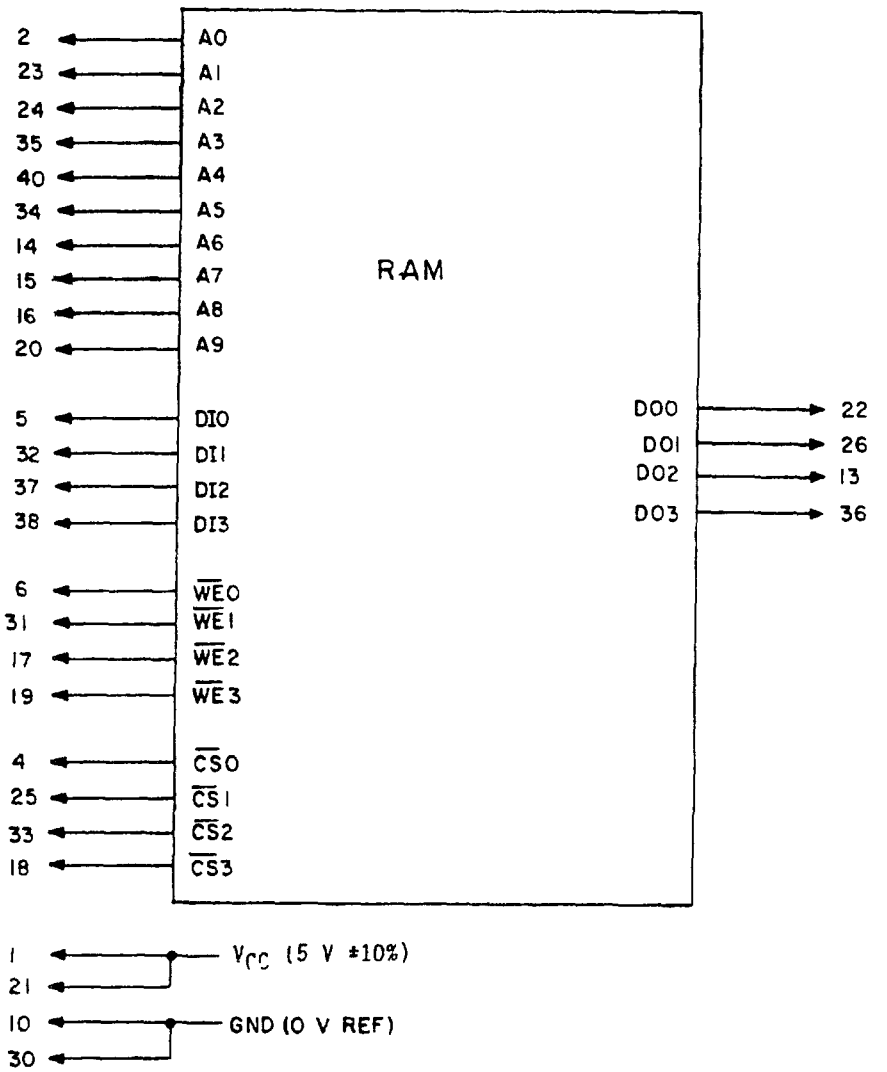
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. A0-A9 = Address inputs.
2. DI0-DI3 = Data inputs.
3. DO0-DO3 = Data outputs.
4. $\overline{WE0}$-$\overline{WE3}$ = Write enable inputs.
5. $\overline{CS0}$-$\overline{CS3}$ = Circuit select inputs.
6. Unused pins: 3, 7, 8, 9, 11, 12, 27, 28, 29, and 39.

FIGURE 1. BYF module function diagram, random access memory.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B4	V_{OUT} (low loaded)	IV	S	0.1	0.9	0.1	0.9	0.1	0.9	0.1	0.9	V
C1-C4	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
C5-C14	Input leakage current	V	X	0	40	0	40	0	40	0	40	µA
C15-C22	Input leakage current	V	X	0	10	0	10	0	10	0	10	µA
D1-D4	Output leakage current, low level outputs	VI	X	-40	40	-40	40	-40	40	-40	40	µA
E1-E4	Output leakage current, high level outputs	VI	X	-40	40	-40	40	-40	40	-40	40	µA
F1-F4	Output short circuit current	VII	X	-30	-70	-30	-70	-30	-70	-30	-70	mA
G1-G4	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
H1-H4	Output voltage, low level	IX	X	0.01	0.40	0.01	0.40	0.01	0.40	0.01	0.40	V
J1-J4	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K4	Input current, high level	X	X	0	50	0	50	0	50	0	50	µA
L1-L4	Input current, low level	XI	X	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	mA
L5-L14	Input current, low level	XI	X	0	-40	0	-40	0	-40	0	-40	µA
L15-L22	Input current, low level	XI	X	0	-10	0	-10	0	-10	0	-10	µA
M1-M4	Address input threshold, low level, check	XII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
M5-M8	Address input threshold, high level, check	XII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
N1,N2	V_{CC} and ground pin continuity	XIII	X	0	1	0	1	0	1	0	1	Ω
N3,N4	V_{CC} and ground pin continuity	XIII	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P	Power supply current	XIV	X	50	180	50	180	50	180	50	180	mA
Q1-Q4	Transition time (t_{TLH}), C_L = 50 pF	XV	S	5	25	5	25	5	25	5	25	ns
Q5-Q8	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	5	40	5	40	5	40	5	40	ns
Q9-Q12	Transition time (t_{THL}), C_L = 50 pF	XV	S	3	15	3	15	3	15	3	15	ns
Q13-Q16	Transition time (t_{THL}), C_L = 150 pF	XV	Q	3	25	3	25	3	25	3	25	ns
R1-R4	Select recovery time (t_{SRL})	XVI	S	5	40	5	40	5	40	5	40	ns
R5-R8	Select recovery time (t_{SRH})	XVI	S	5	40	5	40	5	40	5	40	ns
S1-S8	Dynamic	XVII	X	← No failures →								
S9-S16	Dynamic	XVII	Q	← No failures →								
T1-T8	Input crosstalk, low level	XVIII	Q	← See footnote 4 and table XVIII →								
T9-T16	Input crosstalk, high level	XVIII	Q	← See footnote 4 and table XVIII →								
U	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	mV
V	Power supply filter capacitance	XX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W4	Input capacitance	XXI	Q	2	25	2	25	2	25	2	25	pF
W5-W14	Input capacitance	XXI	Q	2	45	2	45	2	45	2	45	pF
W15-W22	Input capacitance	XXI	Q	2	15	2	15	2	15	2	15	pF
X1-X4	Output capacitance	XXI	Q	2	25	2	25	2	25	2	25	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Initial and end-of-life limits for acceptance requirements and operating temperature are the same.

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4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 6.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 9.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 10. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 6. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through X shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/80-1 supersedes part number M28787/80. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation All outputs at low level	990 mW (outputs not loaded)
Supply voltage	5 V dc ±10 percent
Input voltage Low level High level	 -0.50 V minimum, 0.65 V maximum 2.2 V minimum, V_{CC} maximum

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Input leakage current, A0-A9	
Low level (V_{IL} = 0.0 V)	40 µA maximum
(V_{IL} = 0.4 V)	40 µA maximum
high level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	40 µA maximum
Input leakage current, $\overline{WE}$0-$\overline{WE}$3, DI0-DI3	
Low level (V_{IL} = 0.0 V)	10 µA maximum
(V_{IL} = 0.4 V)	10 µA maximum
High level (V_{IH} = 2.4 V)	10 µA maximum
(V_{IH} = 5.5 V)	10 µA maximum
Input current, $\overline{CS}$0-$\overline{CS}$3	
Low level (V_{IL} = 0.0 V)	-1.73 mA maximum
(V_{IL} = 0.4 V)	-1.6 mA maximum
High level (V_{IH} = 2.4 V)	50 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input clamp diode voltage, $\overline{CS}$0-$\overline{CS}$3	
(V_{CC} = 4.5 V, I_{CLAMP} = -12 mA)	-1.5 V maximum
Output voltage	
Low level (I_{OL} = 16 mA)	0.4 V maximum
High level (I_{OH} = -2 mA)	2.4 V minimum
Output disabled current	
Low level (V_{OL} = 0.4 V)	±40 µA maximum
High level (V_{OH} = 2.4 V)	±40 µA maximum
Output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	16 mA
High level (V_{OH} = 2.4 V)	-2 mA maximum
Input capacitance	
Data in, DI0-DI3	15 pF maximum
Address, A0-A9	45 pF maximum
Circuit select (enable), $\overline{CS}$0-$\overline{CS}$3	25 pF maximum
Write enable, $\overline{WE}$0-$\overline{WE}$3	15 pF maximum
Output capacitance, DO0-DO3	25 pF maximum
Dynamic characteristics (see figure 2)	
Setup time	
Address to write enable (t_{AS})	150 ns minimum
Circuit select (low-to-high) to write enable (high-to-low) to prevent writing ($t_{\overline{SS}}$)	50 ns minimum
Circuit select (high-to-low) to write enable (low-to-high) to assure writing (t_{SS})	300 ns minimum
Data in to write enable (low-to-high) (t_{DS})	300 ns minimum
Hold time	
Write enable (low-to-high) to address (t_{AH})	100 ns minimum
Write enable (low-to-high) to circuit select (t_{SH})	50 ns minimum
Write enable (low-to-high) to data (t_{DH})	100 ns minimum
Write pulse width (t_{WP})	300 ns minimum
Address access time (t_{AA})	
C_L = 50 pF	525 ns maximum
C_L = 150 pF	530 ns maximum

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Select acess time (t_{SA})	
C_L = 50 pF	375 ns maximum
C_L = 150 pF	380 ns maximum
Select recovery time	
Low state to high impedance state (t_{SRL}), $C_L \leq 10$ pF	40 ns maximum
High state to high impedance state (t_{SRH}), $C_L \leq 10$ pF	40 ns maximum
Write recovery time (t_{WR})	
C_L = 50 pF	100 ns maximum
C_L = 150 pF	110 ns maximum

* 6.5 Module function. This module can be used in any of three read/write random access memory configurations:

a. For a 1024-word by 4-bit memory configuration, all four write enable inputs ($\overline{WE}$0-$\overline{WE}$3) and select inputs ($\overline{CS}$0-$\overline{CS}$3) are tied together at the module connector, whereas the four data inputs (DI0-DI3), and the four data outputs (DO0-DO3) are all independent. Functionally, the module is as shown on figure 3.

b. If a 2048-word by 2-bit memory configuration is required, two dual 1024-word by 2-bit groups are formed by pairing circuit selects, write enables, data inputs and data outputs using the procedure outlined below:

Any two of the four circuit select ($\overline{CS}$) signals can be tied together, the remaining two must form the other pair. Then the write enable inputs ($\overline{WE}$), data inputs (DI) and data outputs (DO) of one of the two devices whose $\overline{CS}$s are tied together are paired with the $\overline{WE}$, DI and DO of either of the two devices that form the other pair. Finally, the $\overline{WE}$, DI and DO of the remaining two devices are paired. An example is given below:

Group A pairs	Group B pairs
$\overline{CS}$0 & $\overline{CS}$2	$\overline{CS}$1 & $\overline{CS}$3
$\overline{WE}$0 & $\overline{WE}$3	$\overline{WE}$1 & $\overline{WE}$2
DI0 & DI3	DI1 & DI2
DO0 & DO3	DO1 & DO2

Functionally, the module is as shown on figure 4.

c. A 4096-word by 1-bit memory is achieved by tying together the four write enable inputs ($\overline{WE}$), the four data inputs (DI), and four data outputs (DO), respectively. The four circuit select inputs remain independent. Functionally, the module is as shown on figure 5.

6.5.1 Input to output relationship. The input to output relationships for read and write operations are described below and in table II.

6.5.1.1 Read operation. The memory is addressed by the address inputs (A0-A9) which are connected in parallel to all four RAM circuits and which decode one of the 1024 storage cells in each of the four devices. When the appropriate circuit selects ($\overline{CS}$0-$\overline{CS}$3) go to the low level, data stored in the selected locations of the enabled circuits is read out on the data out pins. Each of the four data outs is buffered by a three state gate on the module. The data read out has the same polarity as the data written in during the previous write cycle.

6.5.1.2 Write operation. The memory is addressed by the A0-A9 inputs which are connected in parallel to all four RAM devices and which decode to one of the 1024 storage cells in each of the four devices. The circuits which are to be written into are enabled by the appropriate low going circuit select and write enable inputs. The data present on the data input pins of the enabled circuits is written into the selected location.

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6.5.2 Paralleling inputs and outputs. In addition to the paralleling of signals on a single module already discussed in 6.5, bussing of like signals between modules is allowed to achieve expanded memory capability. For example, the address terms, data in, write enable, and data out signals of two modules may be connected together respectively to form an 8192-word by 1-bit configuration. A number of memory outputs may be connected to a common bus. All of the common outputs except one are placed in the high impedance state and one output is selected. Two or more common outputs may not be selected simultaneously.

TABLE II. Input to output relationship for each RAM. 1/ - 4/

Inputs			Output	Mode
$\overline{CS}$	$\overline{WE}$	DI	DO	
H	X	X	High Z	Disabled
L	L	H	Undefined	Write high
L	L	L	Undefined	Write low
L	H	X	Data in	Read data from addressed location

1/ H = High level.
2/ L = Low level.
3/ X = Don't care.
4/ High Z = High impedance state.

Agent:
DLA - ES

Preparing activity:
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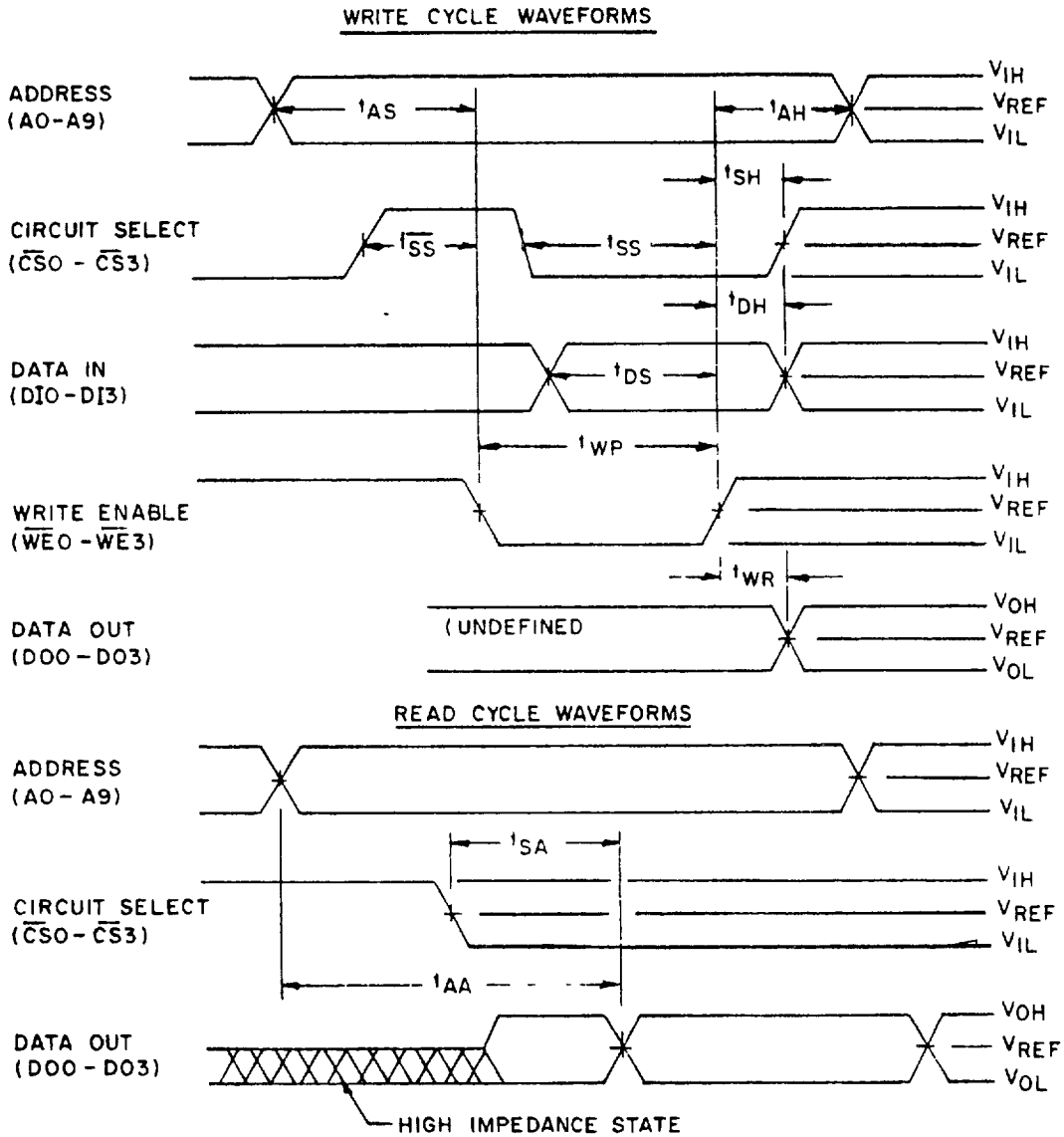


FIGURE 2. Dynamic characteristics waveforms.

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ENABLE/DISABLE CYCLE WAVEFORMS

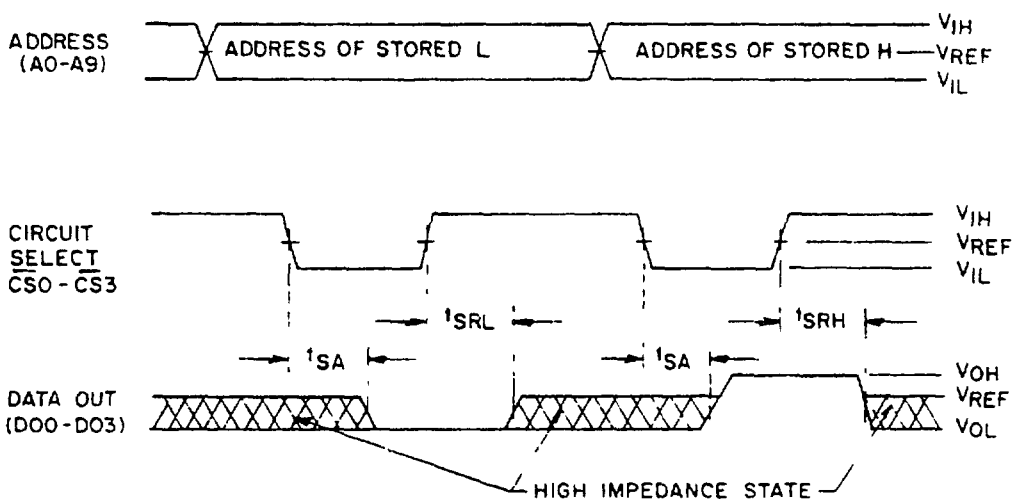


FIGURE 2. Dynamic characteristics waveforms - Continued.

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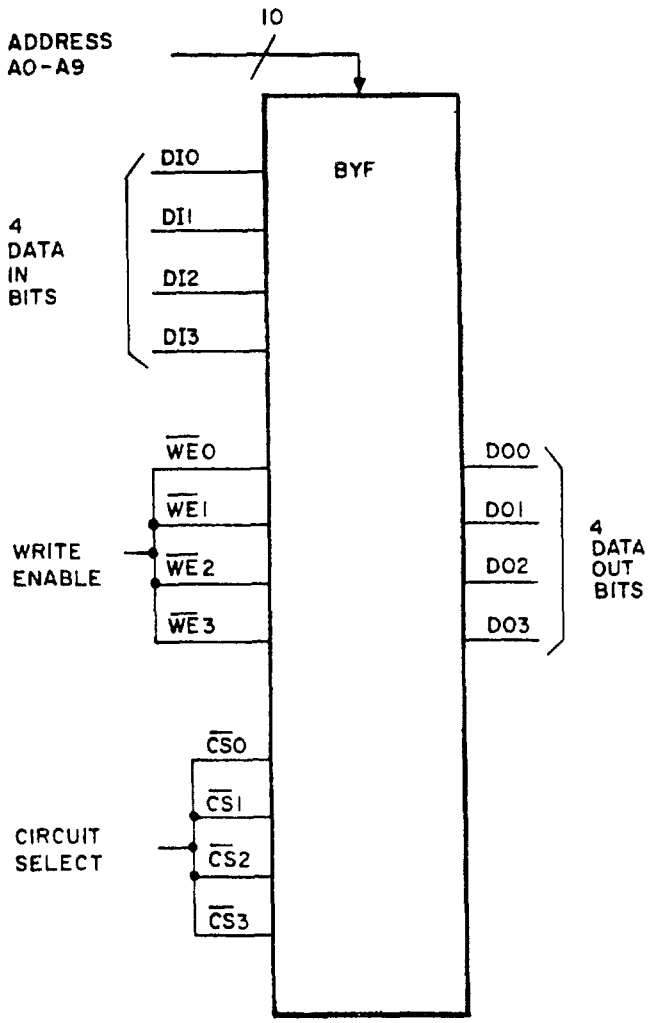


FIGURE 3. 1,024-word by 4-bit configuration.

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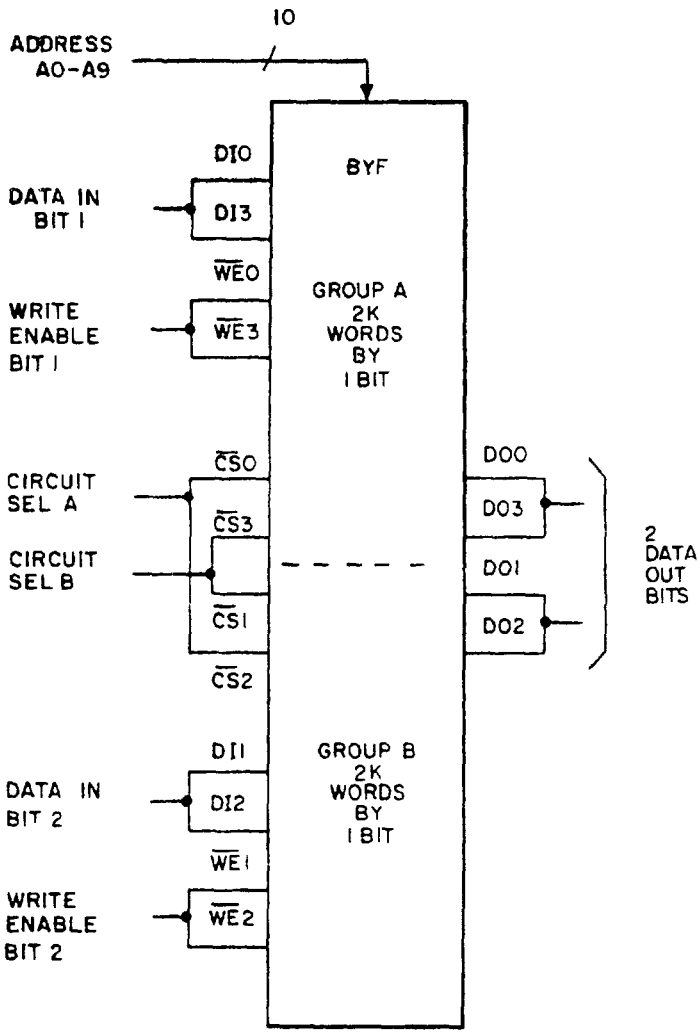


FIGURE 4. 2,048-word by 2-bit configuration.

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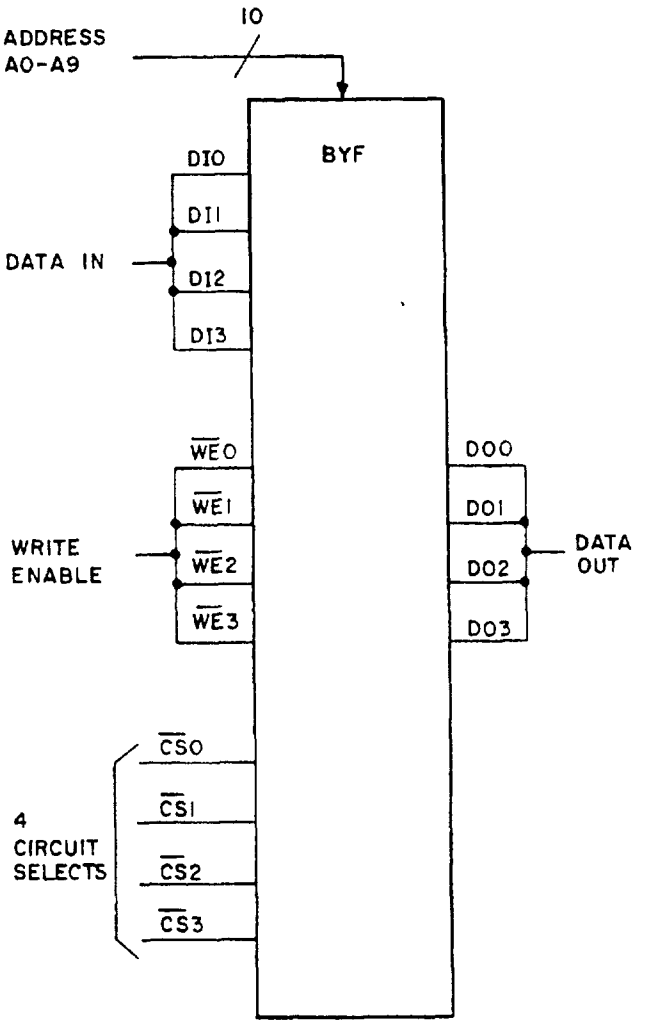


FIGURE 5. 4,096-word by 1-bit configuration.

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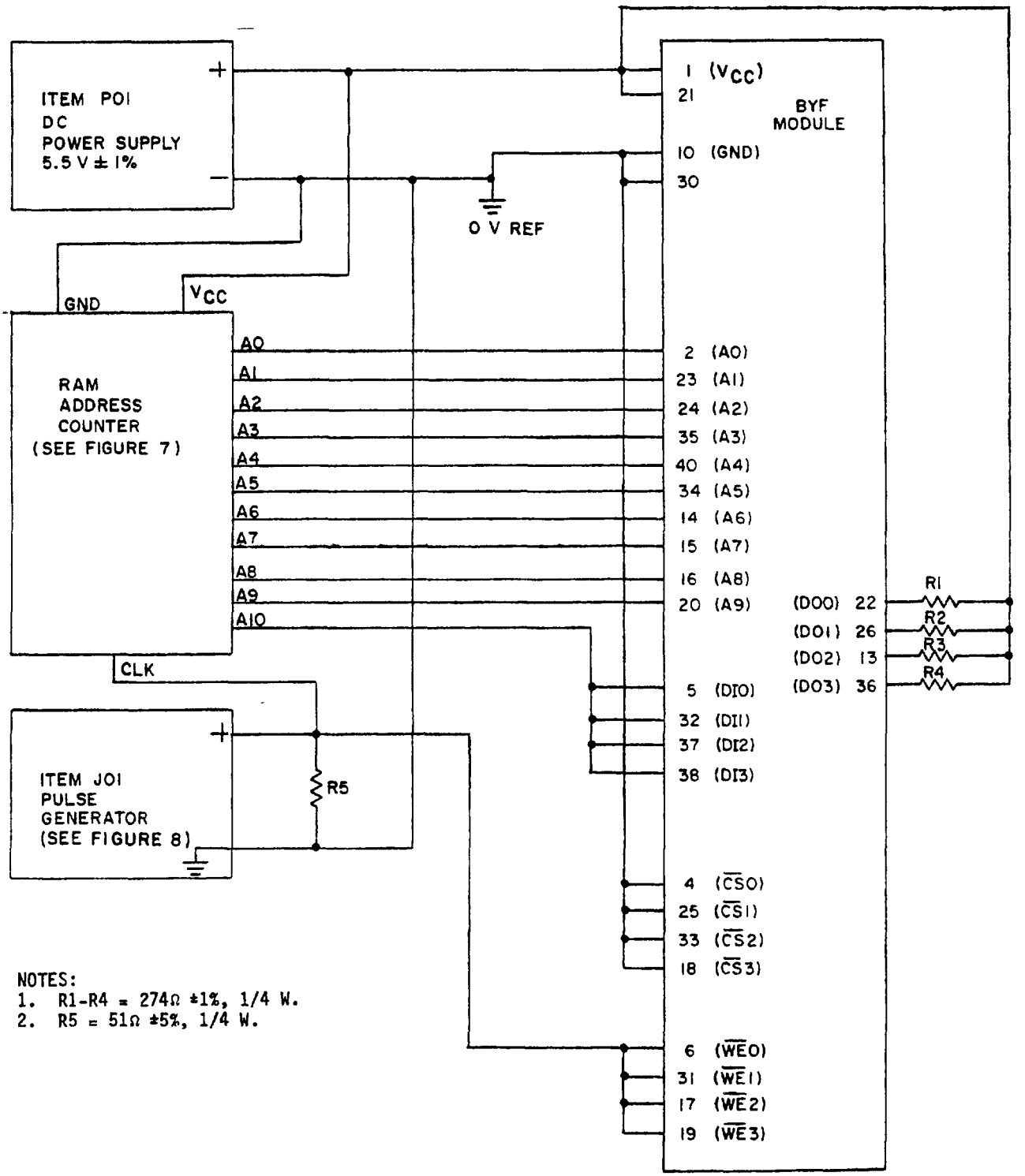
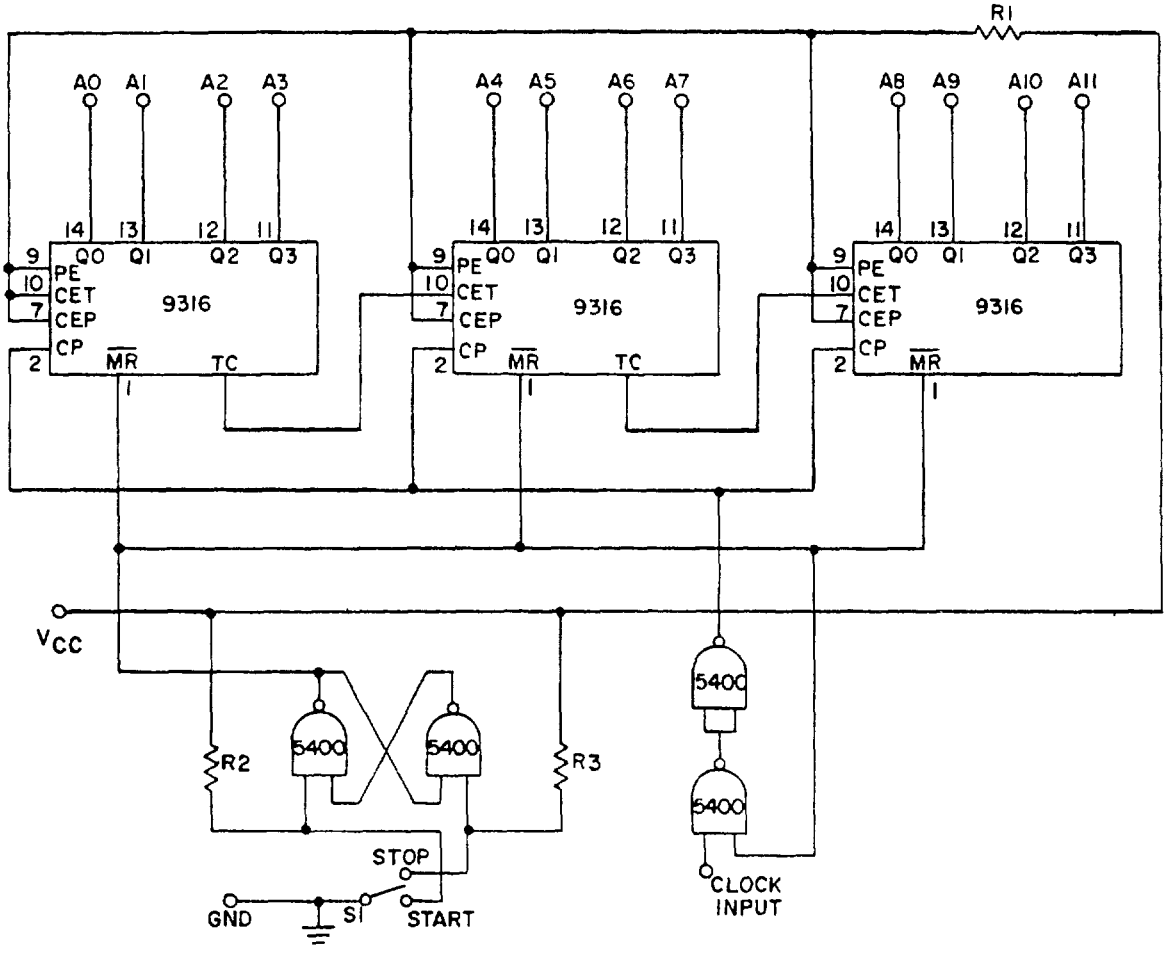


FIGURE 6. Operating temperature and life test circuit.

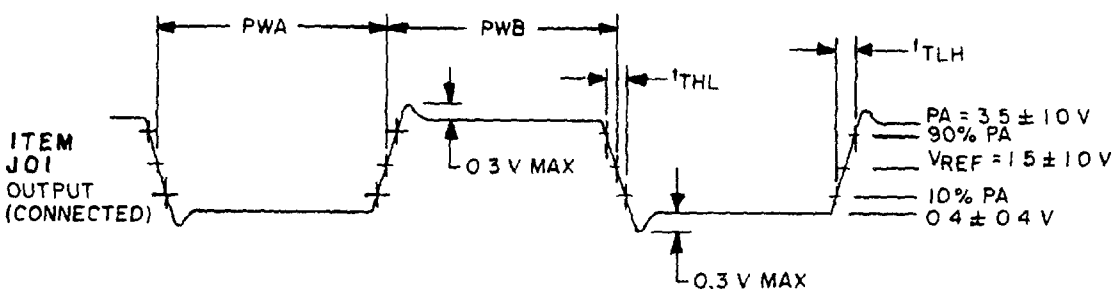
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NOTES:
1. R1-R3 = 2.2 kΩ ±5%, 1/8 W.
2. 3 each 9316 high speed synchronous 4-bit binary counters.
3. 1 each 5400 quad 2-input NAND gate.
4. S1 = SPDT switch.

FIGURE 7. RAM address counter.

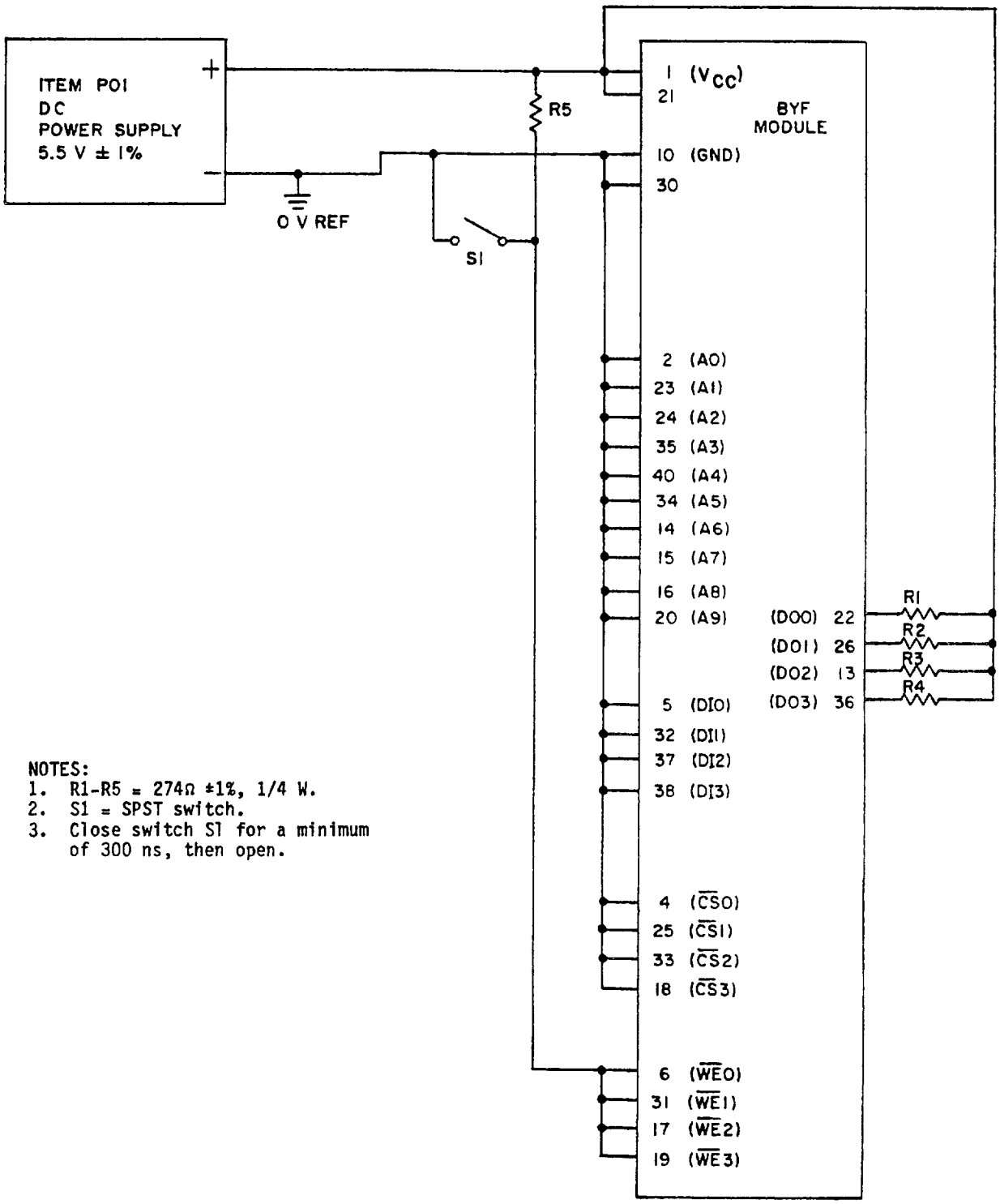
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NOTE: Item J01 output parameters.
PWA = 250 ±20 ns.
PWB = 300 ±30 ns.
$t_{TLH} = t_{THL} = 30 \pm 20$ ns.

FIGURE 8. Operating temperature and life test circuit waveform .

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NOTES:
1. R1-R5 = 274Ω ±1%, 1/4 W.
2. S1 = SPST switch.
3. Close switch S1 for a minimum of 300 ns, then open.

FIGURE 9. Component temperature test circuit.

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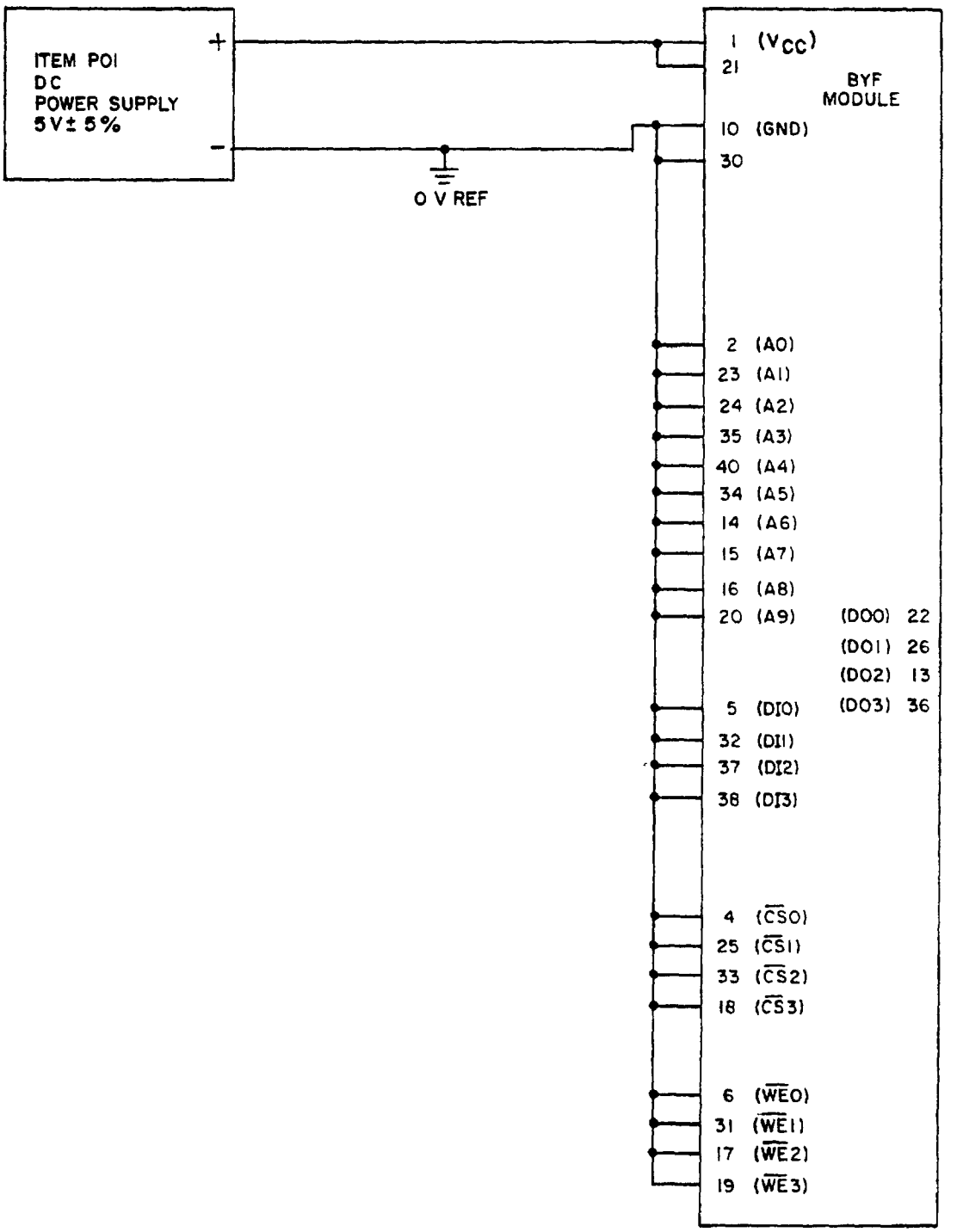


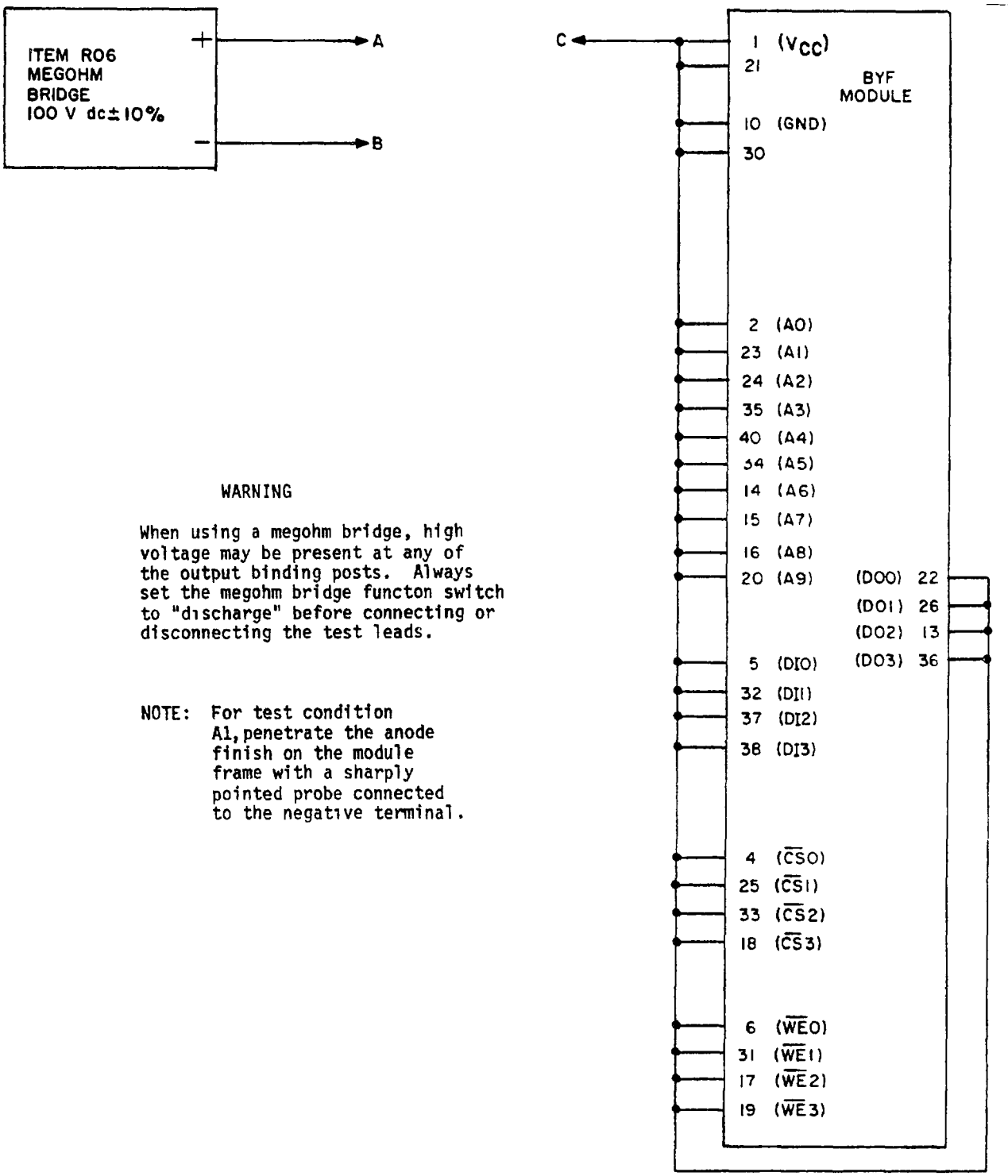
FIGURE 10. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C,3,7,8,9,11, 12,27,28,29,39	Frame	10	∞	MΩ	11
A2	"	C	3,7,8,9,11,12, 27,28,29,39	"	"	"	"
A3	"	3	7,8,9,11,12, 27,28,29,39	"	"	"	"
A4	"	7	3,8,9,11,12, 27,28,29,39	"	"	"	"
A5	"	8	3,7,9,11,12, 27,28,29,39	"	"	"	"
A6	"	9	3,7,8,11,12, 27,28,29,39	"	"	"	"
A7	"	11	3,7,8,9,12, 27,28,29,39	"	"	"	"
A8	"	12	3,7,8,9,11, 27,28,29,39	"	"	"	"
A9	"	27	3,7,8,9,11, 12,28,29,39	"	"	"	"
A10	"	28	3,7,8,9,11, 12,27,29,39	"	"	"	"
A11	"	29	3,7,8,9,11, 12,27,28,39	"	"	"	"
A12	"	39	3,7,8,9,11, 12,27,28,29	"	"	"	"

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge functon switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anode finish on the module frame with a sharply pointed probe connected to the negative terminal.

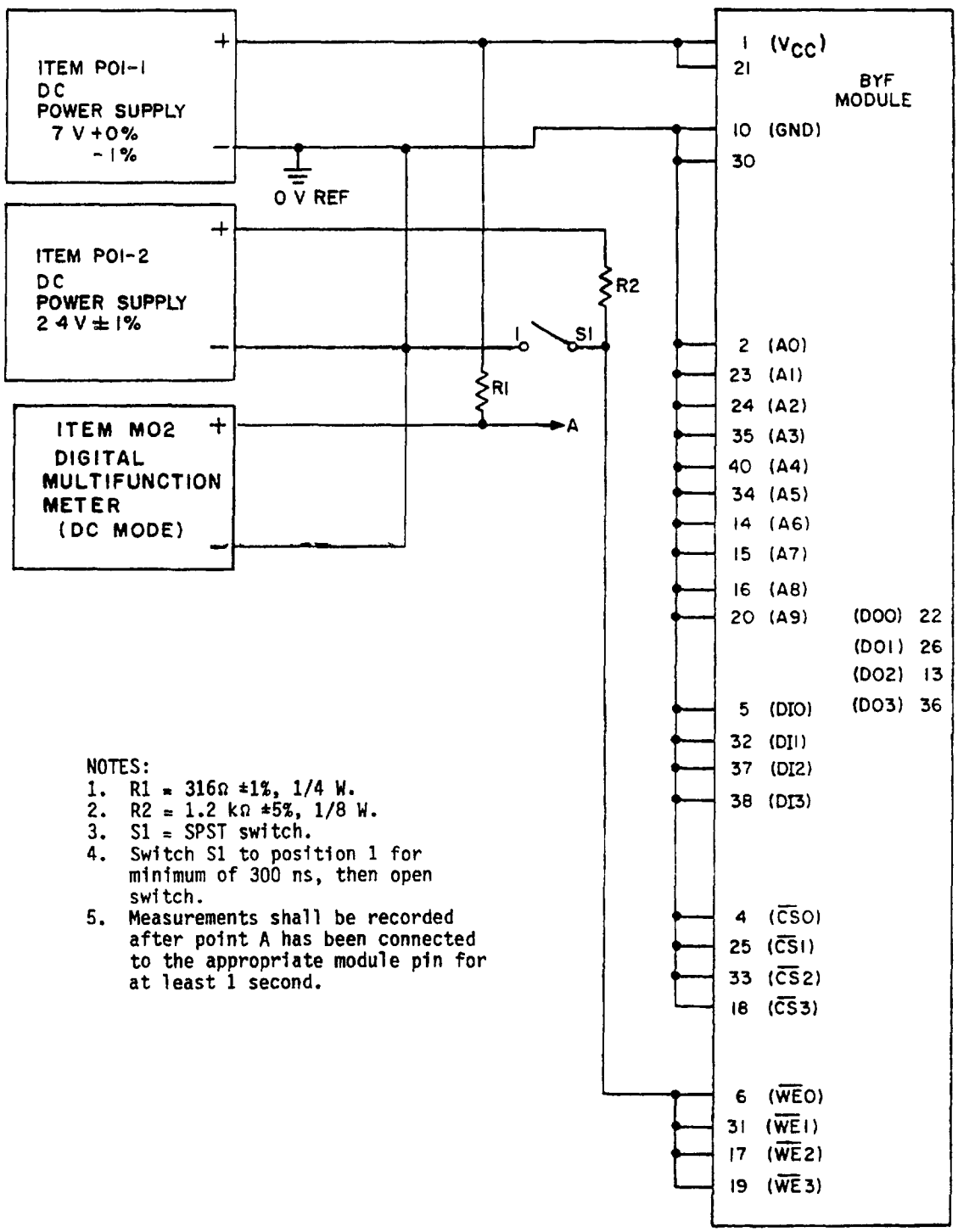
FIGURE 11. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	D00 V_{OUT} (low loaded)	22	0.1	0.9	V	12
B2	D01 "	26	"	"	"	"
B3	D02 "	13	"	"	"	"
B4	D03 "	36	"	"	"	"

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NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. R2 = 1.2 kΩ ±5%, 1/8 W.
3. S1 = SPST switch.
4. Switch S1 to position 1 for minimum of 300 ns, then open switch.
5. Measurements shall be recorded after point A has been connected to the appropriate module pin for at least 1 second.

*FIGURE 12. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current and input leakage current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	$\overline{CS0}$	Input breakdown current	4	2,5,6,14,15,16,17, 18,19,20,23,24,25, 31,32,33,34,35,37, 38,40	0	1	mA	13
C2	$\overline{CS1}$	"	25	2,4,5,6,14,15,16, 17,18,19,20,23,24, 31,32,33,34,35,37, 38,40	"	"	"	"
C3	$\overline{CS2}$	"	33	2,4,5,6,14,15,16, 17,18,19,20,23,24, 25,31,32,34,35,37, 38,40	"	"	"	"
C4	$\overline{CS3}$	"	18	2,4,5,6,14,15,16, 17,19,20,23,24,25, 31,32,33,34,35,37, 38,40	"	"	"	"
C5	A0	Input leakage current	2	4,5,6,14,15,16,17, 18,19,20,23,24,25, 31,32,33,34,35,37, 38,40	0	40	µA	"
C6	A1	"	23	2,4,5,6,14,15,16, 17,18,19,20,24,25, 31,32,33,34,35,37, 38,40	"	"	"	"
C7	A2	"	24	2,4,5,6,14,15,16, 17,18,19,20,23,25, 31,32,33,34,35,37, 38,40	"	"	"	"
C8	A3	"	35	2,4,5,6,14,15,16, 17,18,19,20,23,24, 25,31,32,33,34,37, 38,40	"	"	"	"
C9	A4	"	40	2,4,5,6,14,15,16, 17,18,19,20,23,24, 25,31,32,33,34,35, 37,38	"	"	"	"

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TABLE V. Input breakdown current and input leakage current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C10	A5	Input leakage current	34	2,4,5,6,14,15,16, 17,18,19,20,23,24, 25,31,32,33,35,37, 38,40	0	40	µA	13
C11	A6	"	14	2,4,5,6,15,16,17, 18,19,20,23,24,25, 31,32,33,34,35,37, 38,40	"	"	"	"
C12	A7	"	15	2,4,5,6,14,16,17, 18,19,20,23,24,25, 31,32,33,34,35,37, 38,40	"	"	"	"
C13	A8	"	16	2,4,5,6,14,15,17, 18,19,20,23,24,25, 31,32,33,34,35,37, 38,40	"	"	"	"
C14	A9	"	20	2,4,5,6,14,15,16, 17,18,19,23,24,25, 31,32,33,34,35,37, 38,40	"	"	"	"
C15	$\overline{WE0}$	"	6	2,4,5,14,15,16,17, 18,19,20,23,24,25, 31,32,33,34,35,37, 38,40	"	10	"	"
C16	$\overline{WE1}$	"	31	2,4,5,6,14,15,16, 17,18,19,20,23,24, 25,32,33,34,35,37, 38,40	"	"	"	"
C17	$\overline{WE2}$	"	17	2,4,5,6,14,15,16, 18,19,20,23,24,25, 31,32,33,34,35,37, 38,40	"	"	"	"
C18	$\overline{WE3}$	"	19	2,4,5,6,14,15,16, 17,18,20,23,24,25, 31,32,33,34,35,37, 38,40	"	"	"	"

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TABLE V. Input breakdown current and input leakage current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C19	DIO Input leakage current	5	2,4,6,14,15,16,17, 18,19,20,23,24,25, 31,32,33,34,35,37, 38,40	0	10	µA	13
C20	DI1 "	32	2,4,5,6,14,15,16, 17,18,19,20,23,24, 25,31,33,34,35,37, 38,40	"	"	"	"
C21	DI2 "	37	2,4,5,6,14,15,16, 17,18,19,20,23,24, 25,31,32,33,34,35, 38,40	"	"	"	"
C22	DI3 "	38	2,4,5,6,14,15,16, 17,18,19,20,23,24, 25,31,32,33,34,35, 37,40	"	"	"	"

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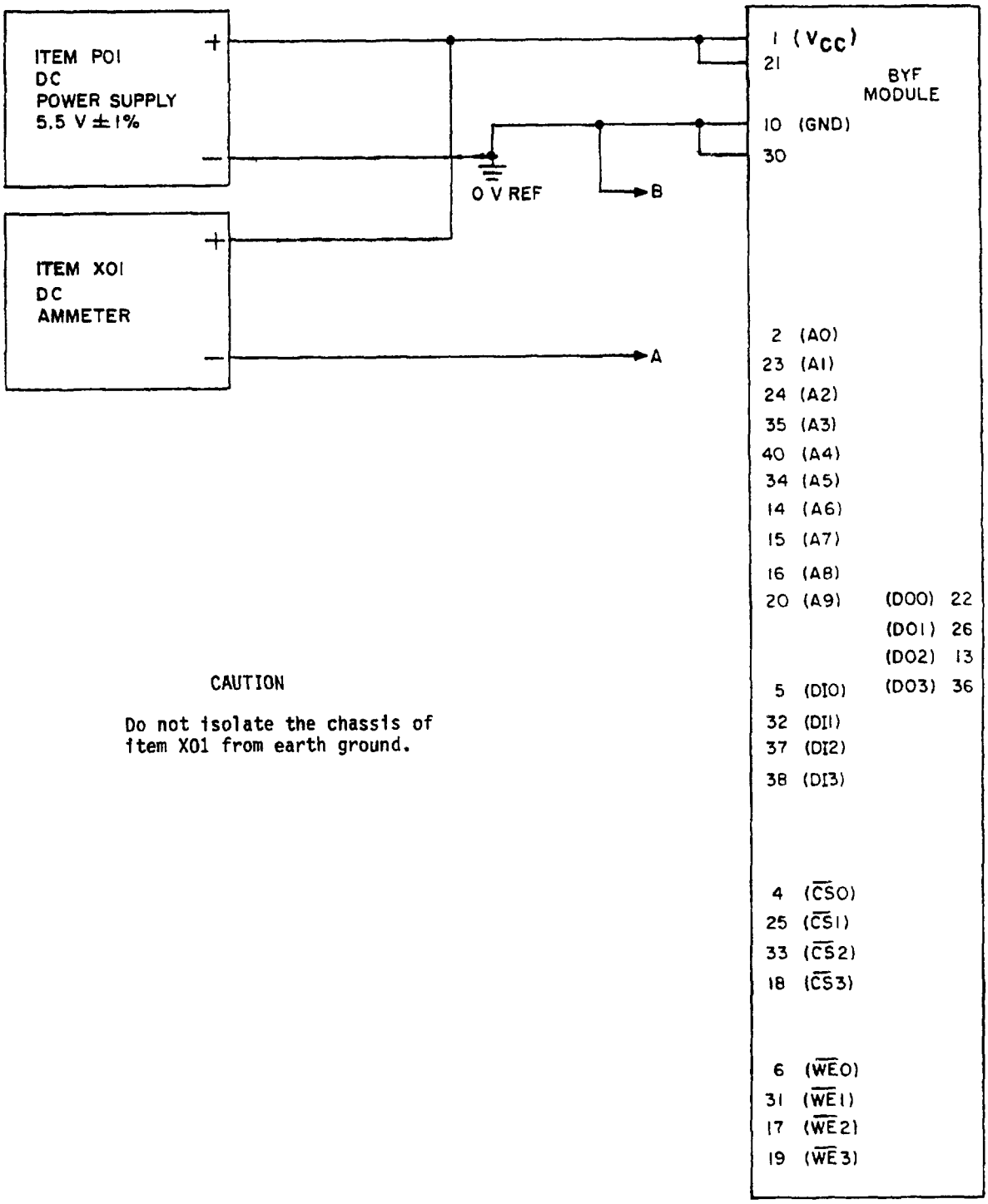


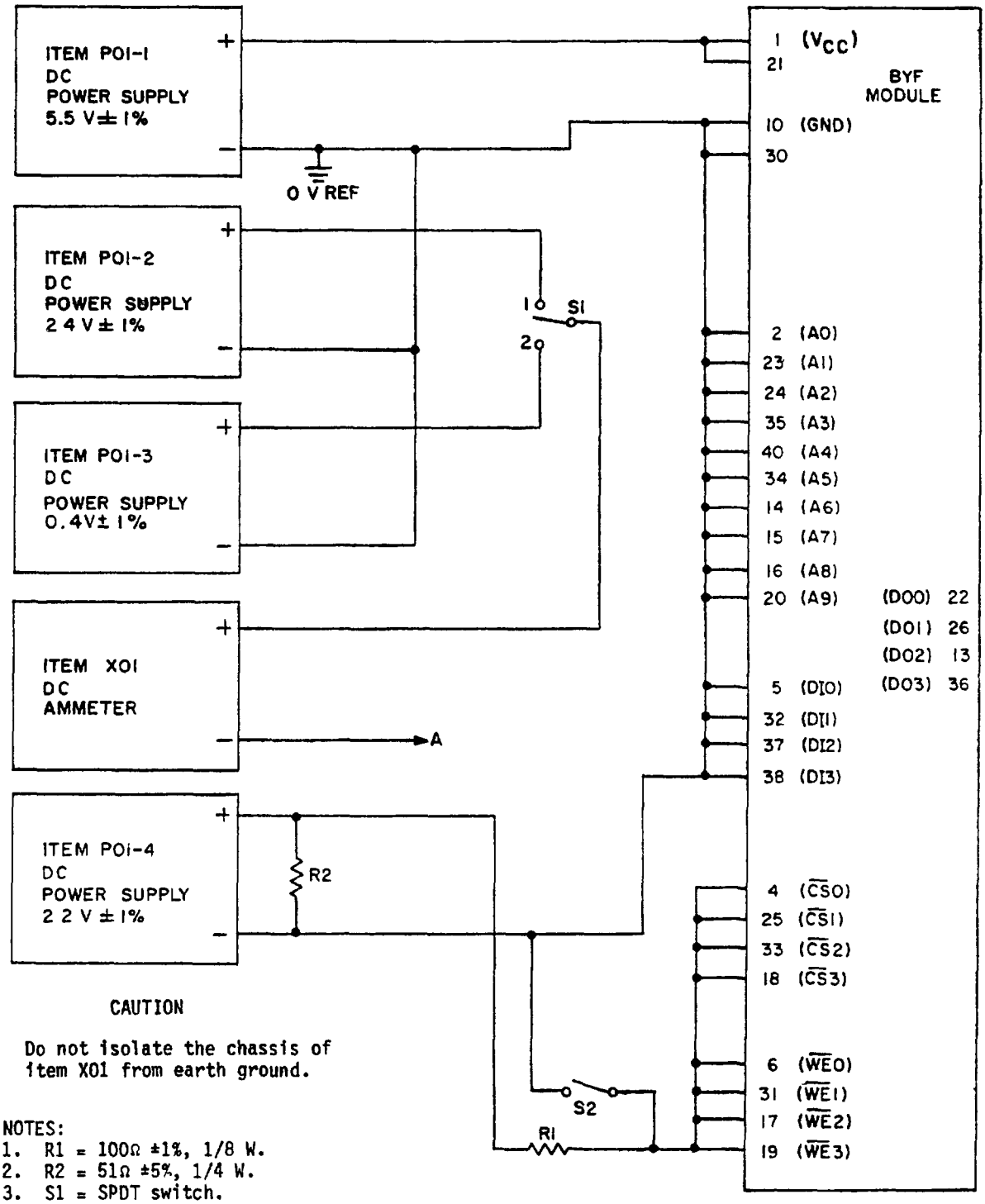
FIGURE 13. Input breakdown current and input leakage current test circuit for table V.

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TABLE VI. Output leakage current (low-level and high-level outputs) test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
D1	DOO	Output leakage current, low level outputs	22	-40	40	µA	14
D2	DO1	"	26	"	"	"	"
D3	DO2	"	13	"	"	"	"
D4	DO3	"	36	"	"	"	"
E1	DOO	Output leakage current, high level outputs	22	"	"	"	"
E2	DO1	"	26	"	"	"	"
E3	DO2	"	13	"	"	"	"
E4	DO3	"	36	"	"	"	"

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NOTES:
1. R1 = 100Ω ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 1/4 W.
3. S1 = SPDT switch.
4. S2 = SPST switch.
5. Switch S1 in position 2 for test conditions D1-D4.
6. Switch S1 in position 1 for test conditions E1-E4.
7. Before performing a test, close switch S2 for a minimum of 500 ns, then open and make measurements.

FIGURE 14. Output leakage current (low-level and high-level outputs) test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	DO0 Output short circuit current	22	-30	-70	mA	15
F2	DO1 "	26	"	"	"	"
F3	DO2 "	13	"	"	"	"
F4	DO3 "	36	"	"	"	"

TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
G1	$\overline{CS}$0 Input clamp diode voltage	4	-0.4	-1.5	V	16
G2	$\overline{CS}$1 "	25	"	"	"	"
G3	$\overline{CS}$2 "	33	"	"	"	"
G4	$\overline{CS}$3 "	18	"	"	"	"

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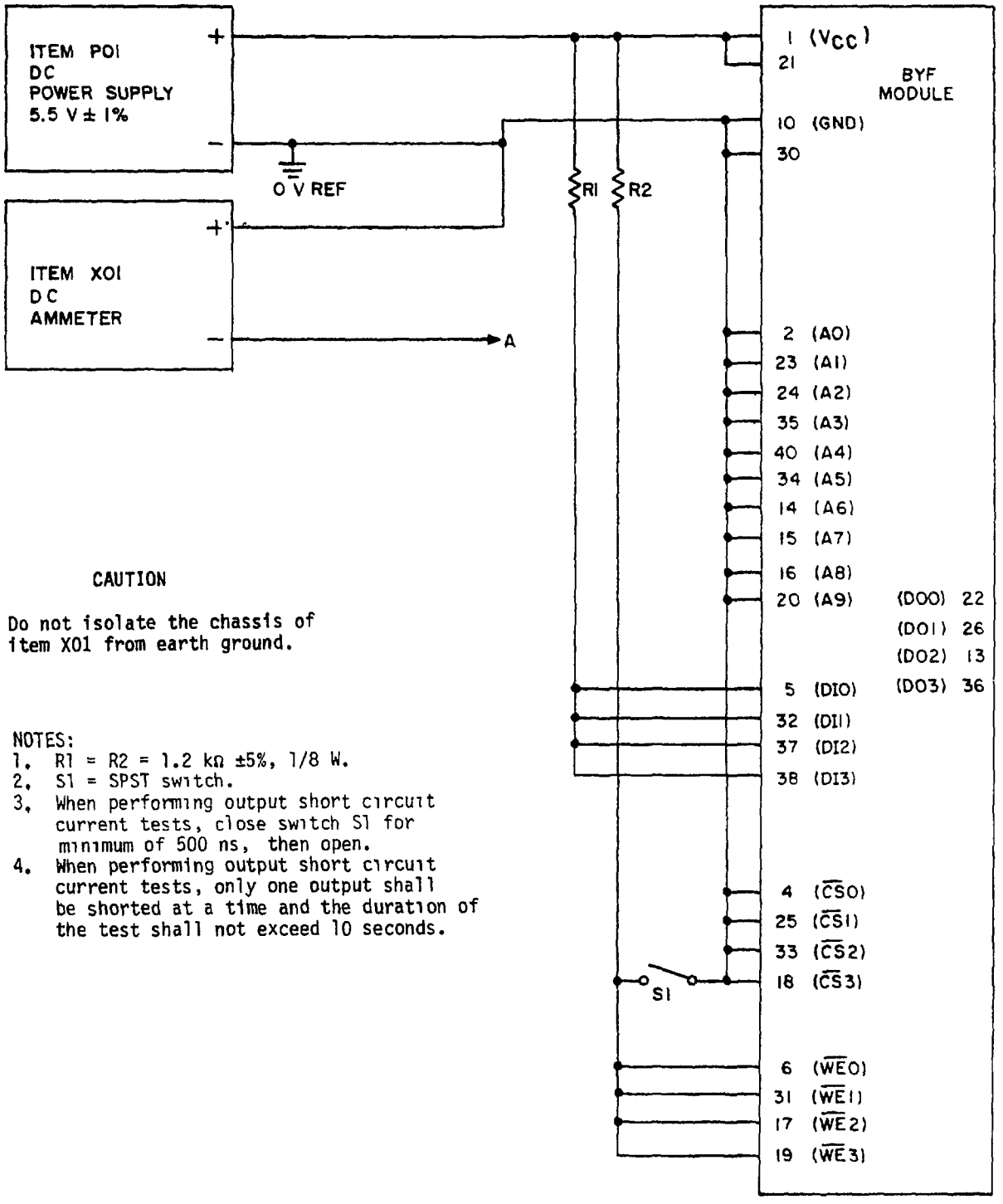
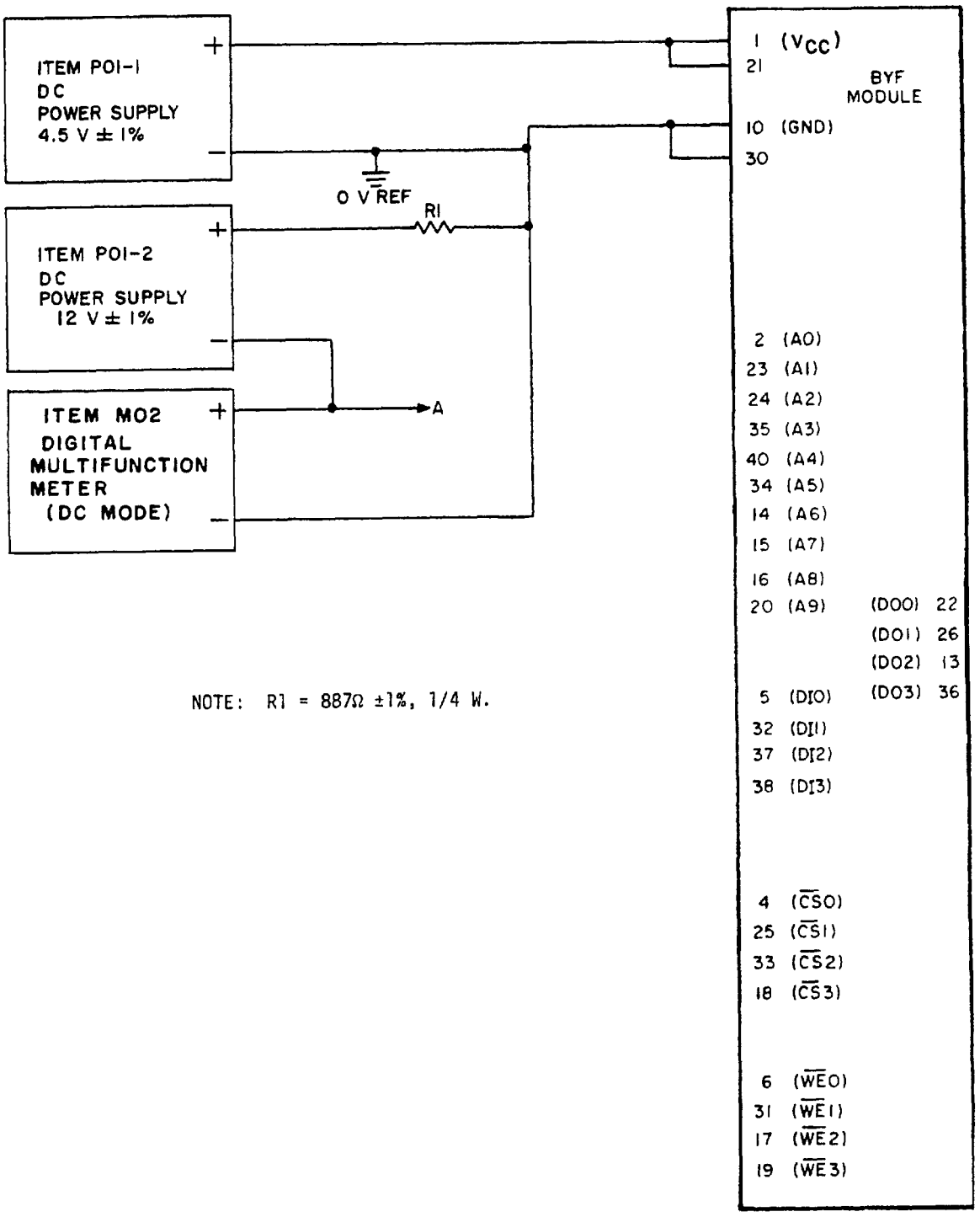


FIGURE 15. Output short circuit current test circuit for table VII.

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NOTE: R1 = 887Ω ±1%, 1/4 W.

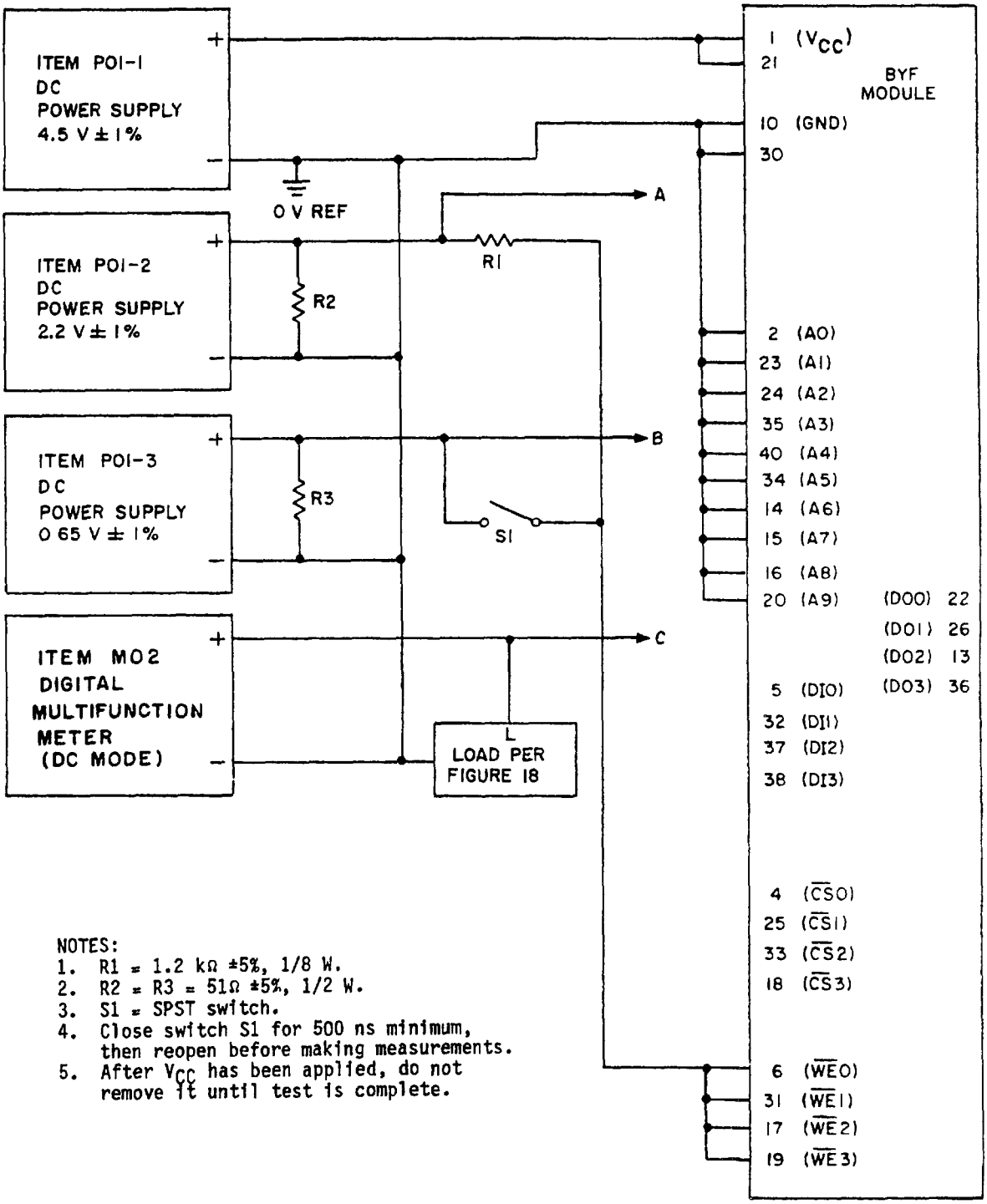
*FIGURE 16. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	D00 Output voltage, low level	Open	4,5,18,25,33	22	0.01	0.40	V	17
H2	D01 "	"	4,18,25,32,33	26	"	"	"	"
H3	D02 "	"	4,18,25,33,37	13	"	"	"	"
H4	D03 "	"	4,18,25,33,38	36	"	"	"	"
J1	D00 Output voltage, high level	5	4,18,25,32,33 37,38	22	2.4	4.5	"	"
J2	D01 "	32	4,5,18,25,33, 37,38	26	"	"	"	"
J3	D02 "	37	4,5,18,25,32, 33,38	13	"	"	"	"
J4	D03 "	38	4,5,18,25,32, 33,37	36	"	"	"	"

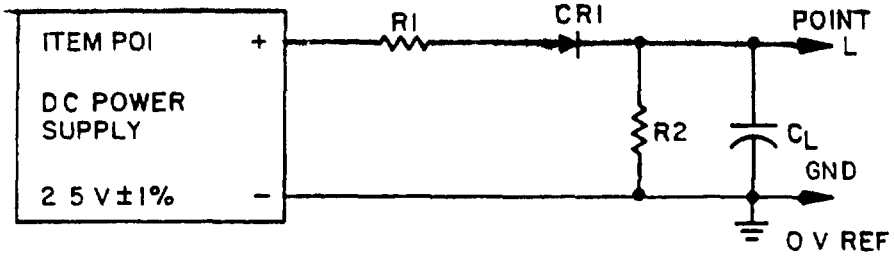
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NOTES:
1. R1 = 1.2 kΩ ±5%, 1/8 W.
2. R2 = R3 = 51Ω ±5%, 1/2 W.
3. S1 = SPST switch.
4. Close switch S1 for 500 ns minimum, then reopen before making measurements.
5. After V_{CC} has been applied, do not remove it until test is complete.

*FIGURE 17. Output voltage (low-level and high-level) test circuit for table IX.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 1.2 kΩ ±1%, 1/8 W.
4. $C_L \leq 10$ pF (including probe and fixture capacitance) for test conditions R1-R8.
5. $C_L = 50$ ±5 pF (including probe and fixture capacitance) for test conditions M1-M8, Q1-Q4, Q9-Q12, S1-S8, and T1-T16..
6. $C_L = 150$ ±5 pF (including probe and fixture capacitance) for test conditions Q5-Q8, Q13-Q16, and S9-S16.
7. Capacitor C_L is not required for test conditions H1-H4 and J1-J4.
8. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

FIGURE 18. BYF logic load circuit.

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TABLE X. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	$\overline{CS0}$ Input current, high level	18,25,33	4	0	50	µA	19
K2	$\overline{CS1}$ "	4,18,33	25	"	"	"	"
K3	$\overline{CS2}$ "	4,18,25	33	"	"	"	"
K4	$\overline{CS3}$ "	4,25,33	18	"	"	"	"

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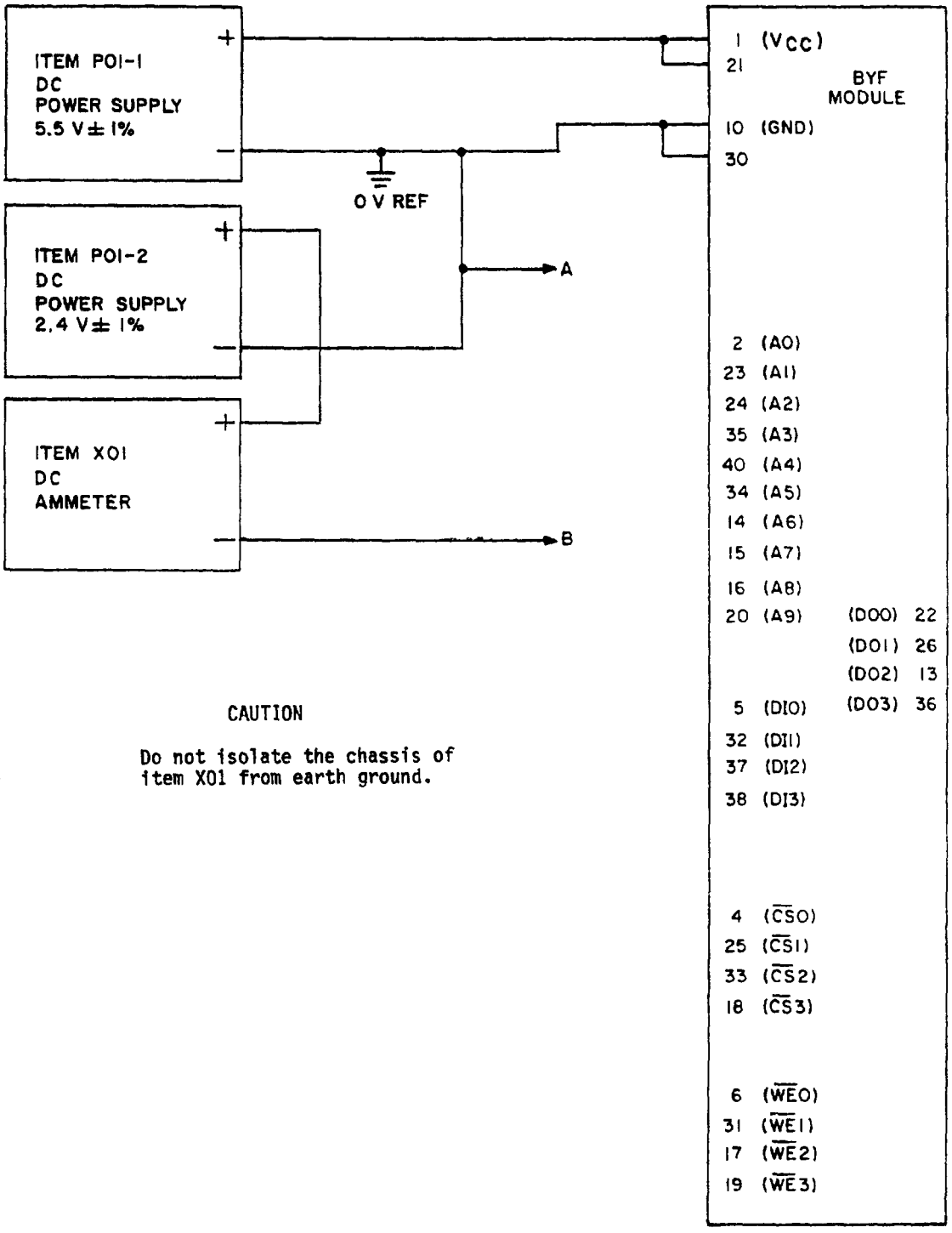


FIGURE 19. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	$\overline{CS}0$ Input current, low level	2,5,6,14,15,16, 17,18,19,20,23, 24,25,31,32,33, 34,35,37,38,40	4	-0.10	-1.73	mA	20
L2	$\overline{CS}1$ "	2,4,5,6,14,15, 16,17,18,19,20, 23,24,31,32,33, 34,35,37,38,40	25	"	"	"	"
L3	$\overline{CS}2$ "	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,32, 34,35,37,38,40	33	"	"	"	"
L4	$\overline{CS}3$ "	2,4,5,6,14,15, 16,17,19,20,23, 24,25,31,32,33, 34,35,37,38,40	18	"	"	"	"
L5	A0 "	4,5,6,14,15,16, 17,18,19,20,23, 24,25,31,32,33, 34,35,37,38,40	2	0	-40	µA	"
L6	A1 "	2,4,5,6,14,15, 16,17,18,19,20, 24,25,31,32,33, 34,35,37,38,40	23	"	"	"	"
L7	A2 "	2,4,5,6,14,15, 16,17,18,19,20, 23,25,31,32,33, 34,35,37,38,40	24	"	"	"	"
L8	A3 "	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,32, 33,34,37,38,40	35	"	"	"	"
L9	A4 "	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,32, 33,34,35,37,38	40	"	"	"	"

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TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L10	A5 Input current, low level	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,32, 33,35,37,38,40	34	0	-40	µA	20
L11	A6 "	2,4,5,6,15,16, 17,18,19,20,23, 24,25,31,32,33, 34,35,37,38,40	14	"	"	"	"
L12	A7 "	2,4,5,6,14,16, 17,18,19,20,23, 24,25,31,32,33, 34,35,37,38,40	15	"	"	"	"
L13	A8 "	2,4,5,6,14,15, 17,18,19,20,23, 24,25,31,32,33, 34,35,37,38,40	16	"	"	"	"
L14	A9 "	2,4,5,6,14,15, 16,17,18,19,23, 24,25,31,32,33, 34,35,37,38,40	20	"	"	"	"
L15	$\overline{WE0}$ "	2,4,5,14,15,16, 17,18,19,20,23, 24,25,31,32,33, 34,35,37,38,40	6	"	-10	"	"
L16	$\overline{WE1}$ "	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,32,33, 34,35,37,38,40	31	"	"	"	"
L17	$\overline{WE2}$ "	2,4,5,6,14,15, 16,18,19,20,23, 24,25,31,32,33, 34,35,37,38,40	17	"	"	"	"
L18	$\overline{WE3}$ "	2,4,5,6,14,15, 16,17,18,20,23, 24,25,31,32,33, 34,35,37,38,40	19	"	"	"	"

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TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L19	DI0	Input current, low level	2,4,6,14,15,16, 17,18,19,20,23, 24,25,31,32,33, 34,35,37,38,40	5	0	-10	µA	20
L20	DI1	"	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,33, 34,35,37,38,40	32	"	"	"	"
L21	DI2	"	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,32, 33,34,35,38,40	37	"	"	"	"
L22	DI3	"	2,4,5,6,14,15, 16,17,18,19,20, 23,24,25,31,32, 33,34,35,37,40	38	"	"	"	"

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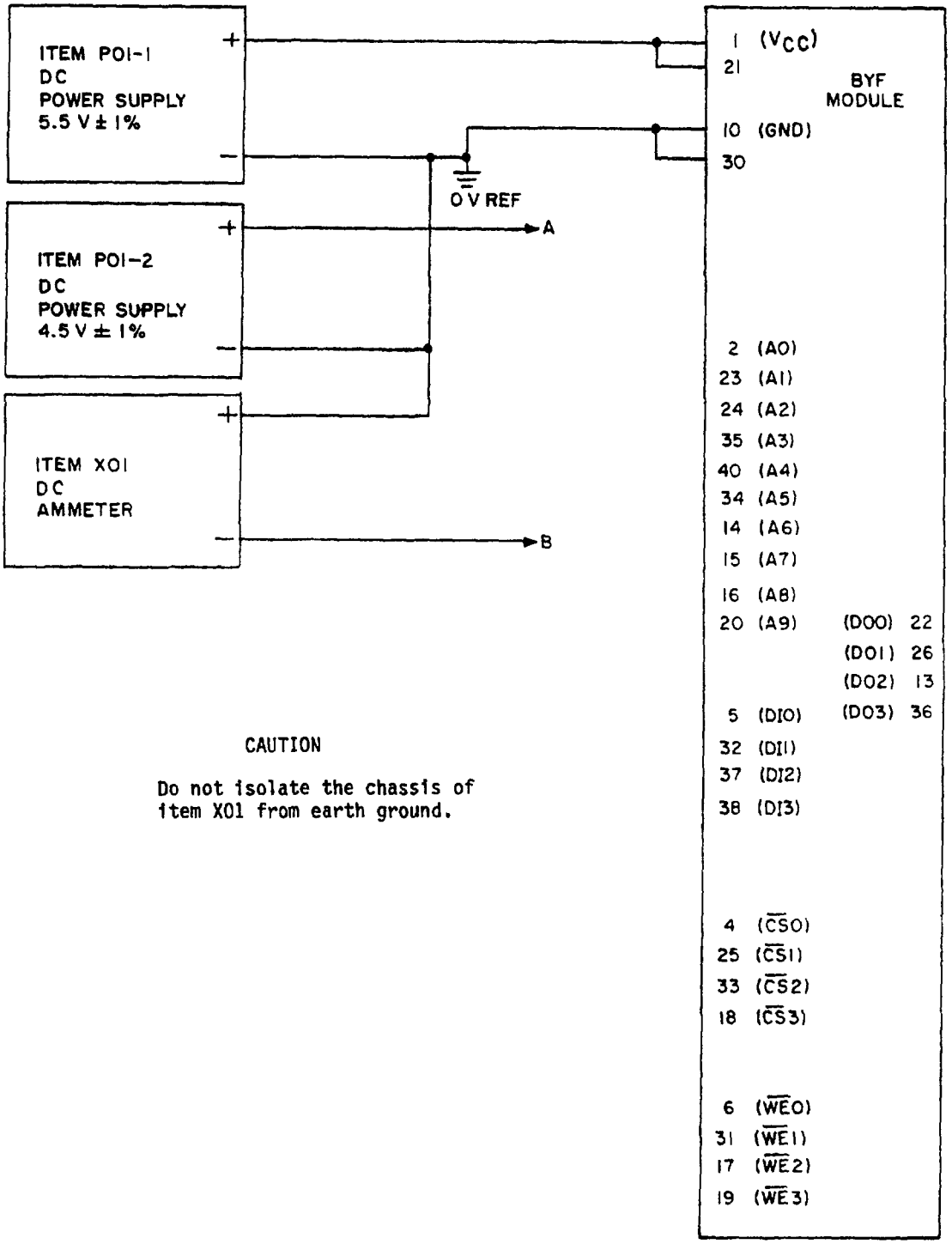


FIGURE 20. Input current (low-level) test circuit for table XI.

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TABLE XII. Address input threshold (low-level and high-level) check test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Address input threshold, low level, check	6,17,19,31	2,14,15,16, 20,23,24, 34,35,40,4, 25,18,33	22	0.0	0.4	V	21
M2	"	"	"	26	"	"	"	"
M3	"	"	"	13	"	"	"	"
M4	"	"	"	36	"	"	"	"
M5	Address input threshold, high level, check	2,6,14,15, 16,17,19, 20,23,24, 31,34,35,40	4,18,25,33	22	2.4	4.5	"	"
M6	"	"	"	26	"	"	"	"
M7	"	"	"	13	"	"	"	"
M8	"	"	"	36	"	"	"	"

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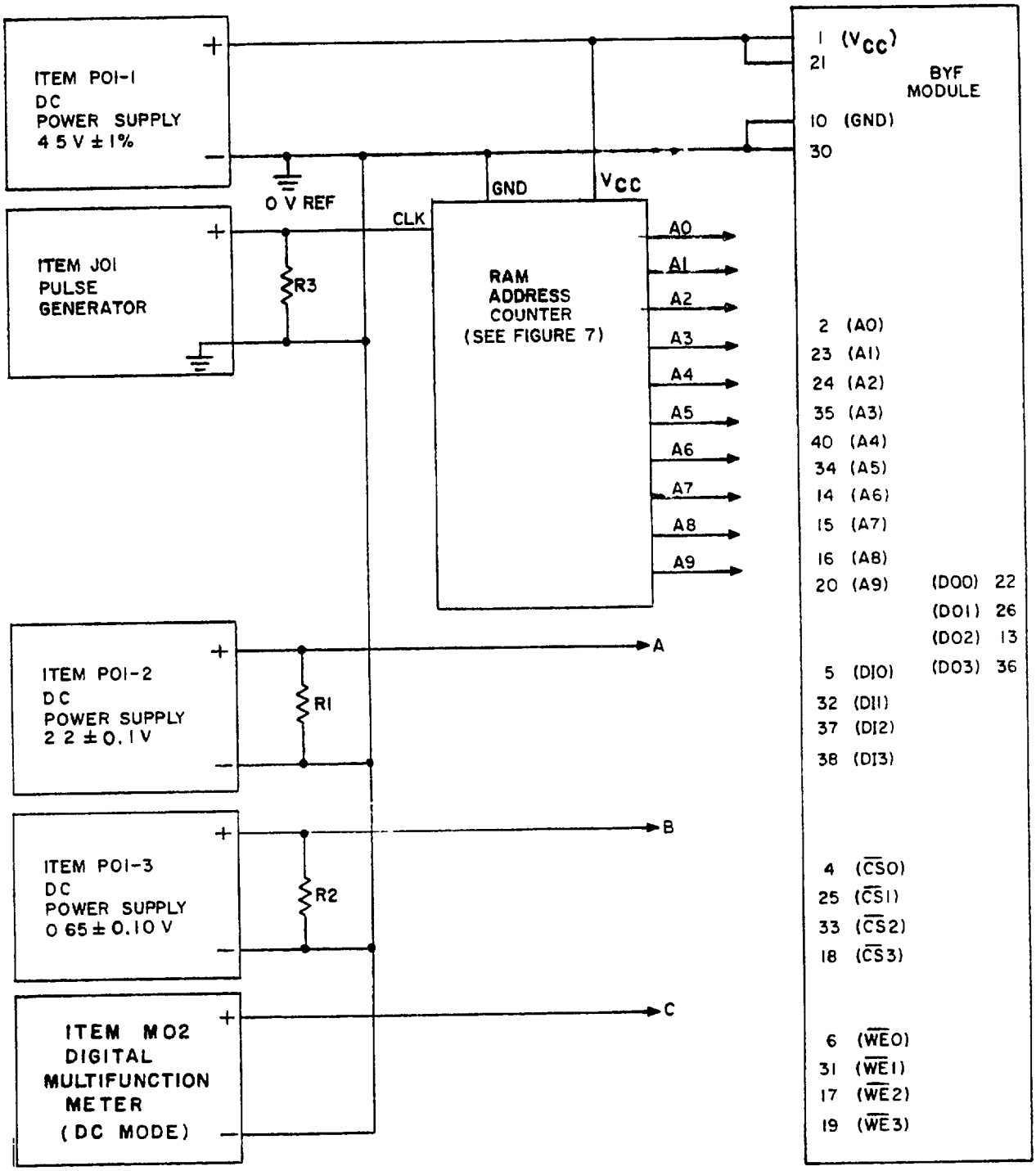


*FIGURE 21. Address input threshold (low-level and high-level) check test circuit for table XII.

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NOTES:
1. R1 = R2 = R3 = 51Ω ±5%, 1 W.
2. Load all outputs with the load as specified on figure 18.
3. Item J01 parameters:
   (See figure 25 for parameter definition)

   a. Amplitude = 4.0 ±0.1 V.
   b. Pulse width = 1,000 ±50 ns.
   c. Pulse repetition period = 2,000 ns ±10%.
   d. $t_{TLH} = t_{THL}$ = 25 ±20 ns.

4. Test sequence for test conditions M1-M4:

   a. Connect RAM address counter to address inputs.
   b. Connect pins 5, 32, 37, and 38 to pin 1. Connect pins 4, 6, 17, 18, 19, 25, 31, and 33 to pin 10.
   c. Initiate RAM address counter for 5 seconds. This loads all memory locations with a high level.
   d. Disconnect pins 6, 17, 19, and 31 from pin 10 and connect to pin 1.
   e. Disconnect the RAM address counter. Connect pins 2, 14, 15, 16, 20, 23, 24, 34, 35, and 40 to pin 10. Connect pins 5, 32, 37, and 38 to pin 10.
   f. Disconnect pins 6, 17, 19, and 31 from pin 1 and connect to pin 10 for a minimum of 500 ns, then connect to point A. Verify that this loads memory location zero with a low level and all other locations with a high level.
   g. Without disconnecting pins 6, 17, 19, and 31 from point A and without interrupting power perform tests as specified in test conditions M1-M4.

5. Test sequence for test conditions M5-M8:

   a. Connect RAM address counter to address inputs.
   b. Connect pins 4, 5, 6, 17, 18, 19, 25, 31, 32, 33, 37, and 38 to pin 10.
   c. Initiate RAM address counter for 5 seconds. This loads all memory locations with a low level.
   d. Disconnect pins 6, 17, 19, and 31 from pin 10 and connect to pin 1.
   e. Disconnect the RAM address counter. Connect pins 2, 14, 15, 16, 20, 23, 24, 34, 35, and 40 to pin 1. Connect pins 5, 32, 37, and 38 to pin 1.
   f. Disconnect pins 6, 17, 19, and 31 from pin 1 and connect to pin 10 for a minimum of 500 ns, then connect to point A. Verify that this loads memory location 1023 with a high level and all other locations with a low level.
   g. Without disconnecting pins 6, 17, 19, and 31 from point A and without interrupting power perform tests as specified in test conditions M5-M8.

FIGURE 21. Address input threshold (low-level and high-level) check test circuit for table XII - Continued.

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TABLE XIII. V_{CC} and ground pin continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	V_{CC} pin continuity	1	21	0	1	Ω	22
N2	Ground pin continuity	10	30	0	1	"	"
N3	V_{CC} pin continuity	1	21	0.0	0.1	"	"
N4	Ground pin continuity	10	30	0.0	0.1	"	"

TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply current	50	180	mA	23

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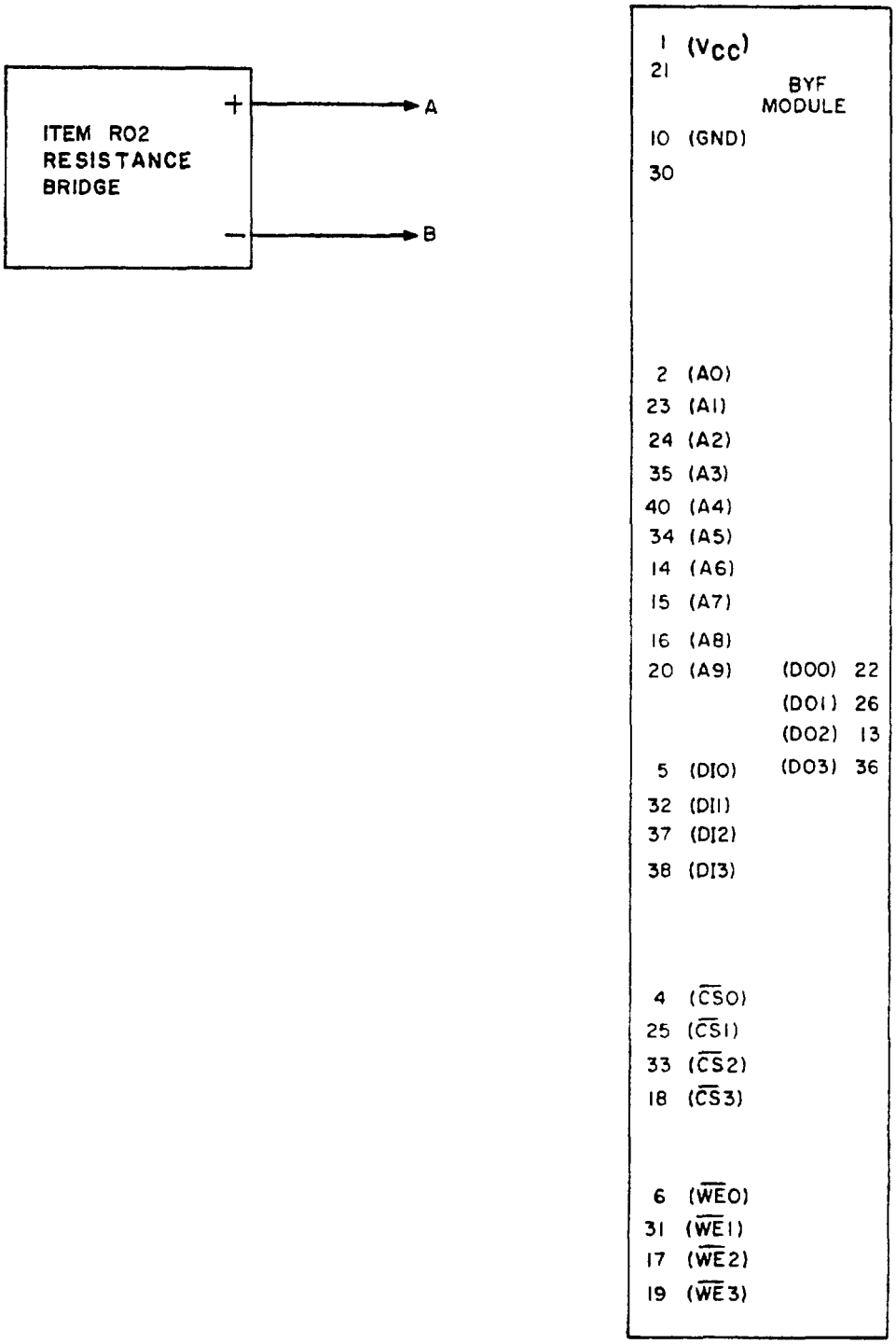


FIGURE 22. V_{CC} and ground pin continuity test circuit for table XIII.

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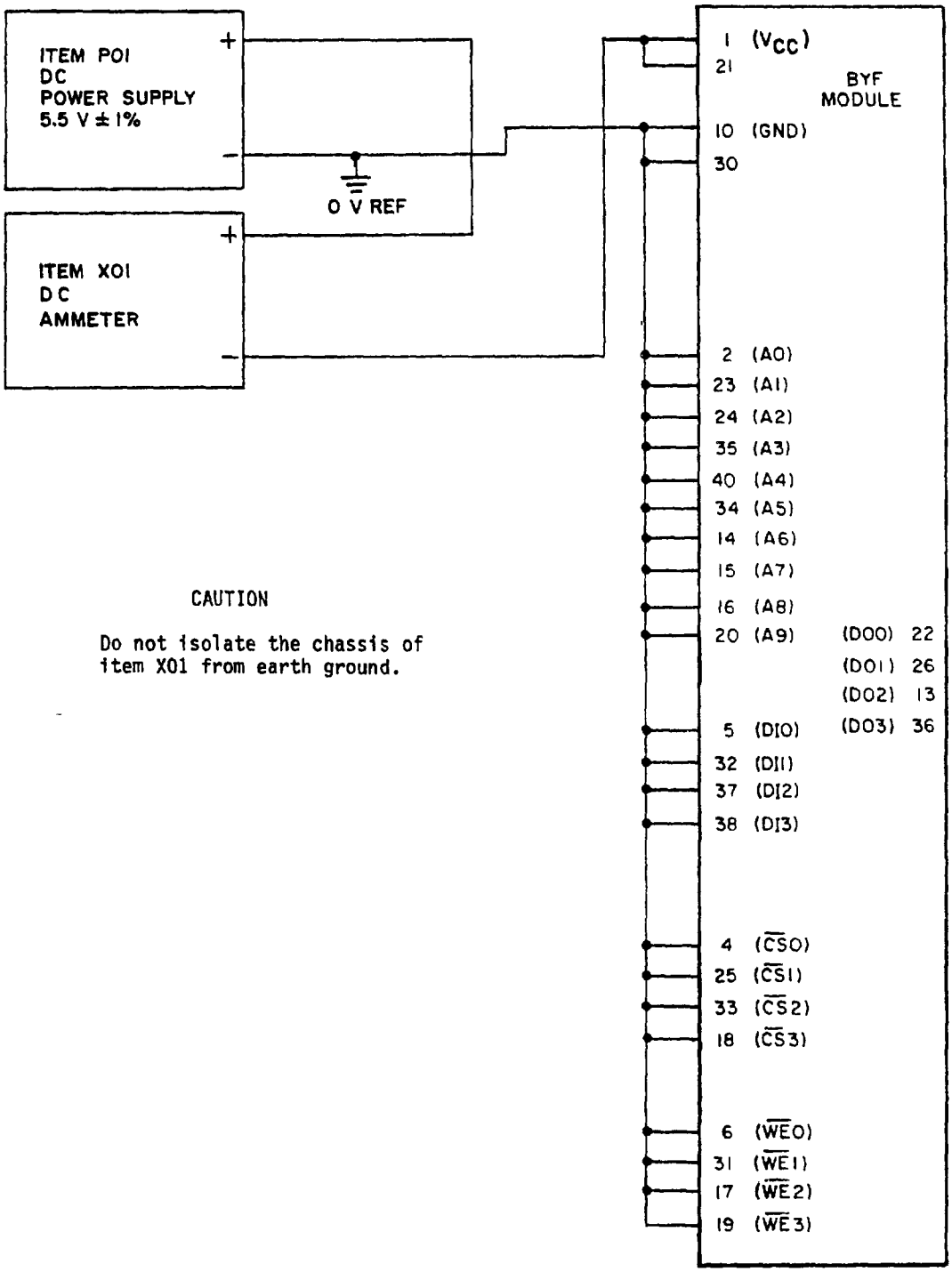


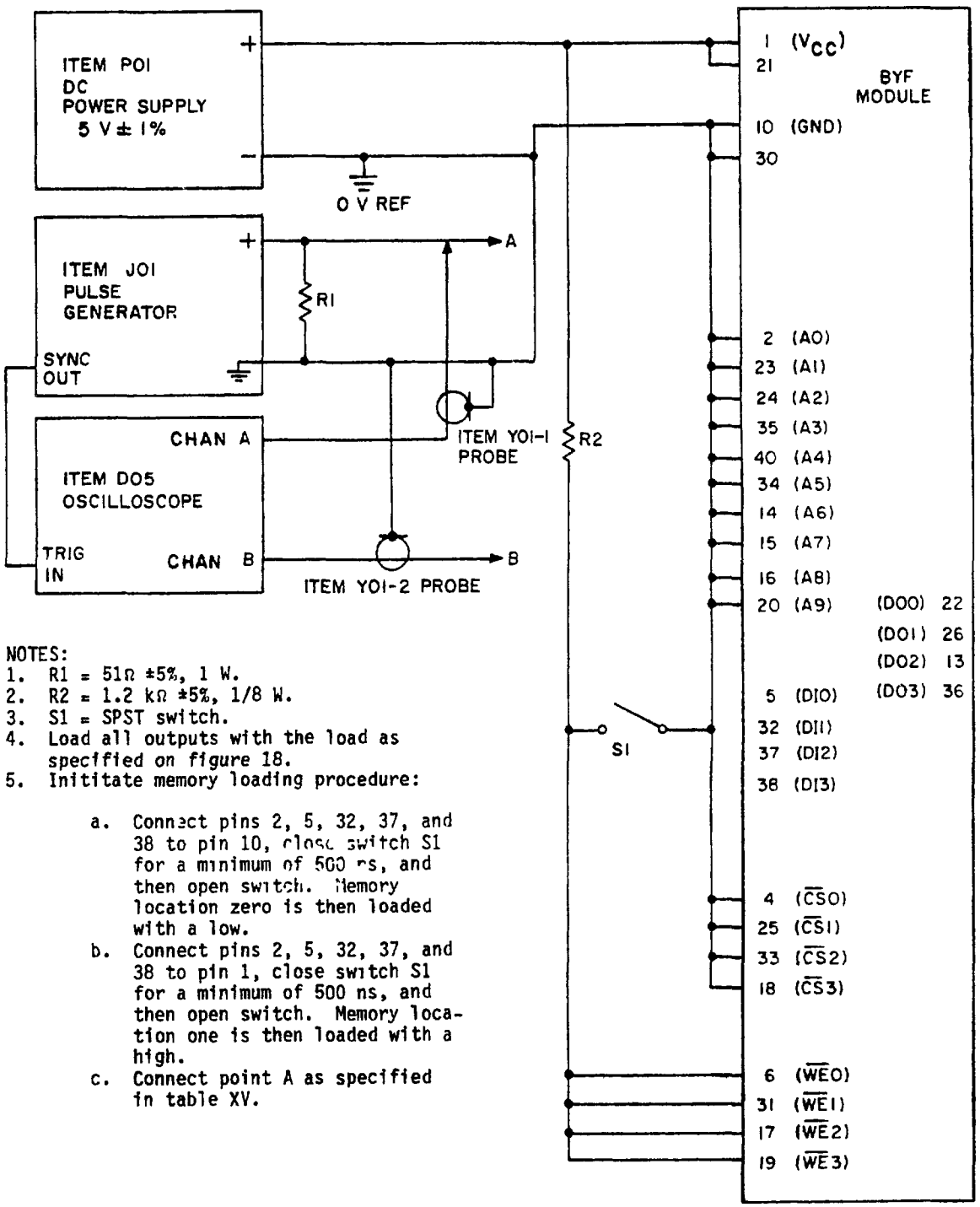
FIGURE 23. Power supply current test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	D00 Transition time (t_{TLH}), C_L = 50 pF	2	22	5	25	ns	24,25
Q2	D01 "	"	26	"	"	"	"
Q3	D02 "	"	13	"	"	"	"
Q4	D03 "	"	36	"	"	"	"
Q5	D00 Transition time (t_{TLH}), C_L = 150 pF	"	22	"	40	"	"
Q6	D01 "	"	26	"	"	"	"
Q7	D02 "	"	13	"	"	"	"
Q8	D03 "	"	36	"	"	"	"
Q9	D00 Transition time (t_{THL}), C_L = 50 pF	"	22	3	15	"	"
Q10	D01 "	"	26	"	"	"	"
Q11	D02 "	"	13	"	"	"	"
Q12	D03 "	"	36	"	"	"	"
Q13	D00 Transition time (t_{THL}), C_L = 150 pF	"	22	"	25	"	"
Q14	D01 "	"	26	"	"	"	"
Q15	D02 "	"	13	"	"	"	"
Q16	D03 "	"	36	"	"	"	"

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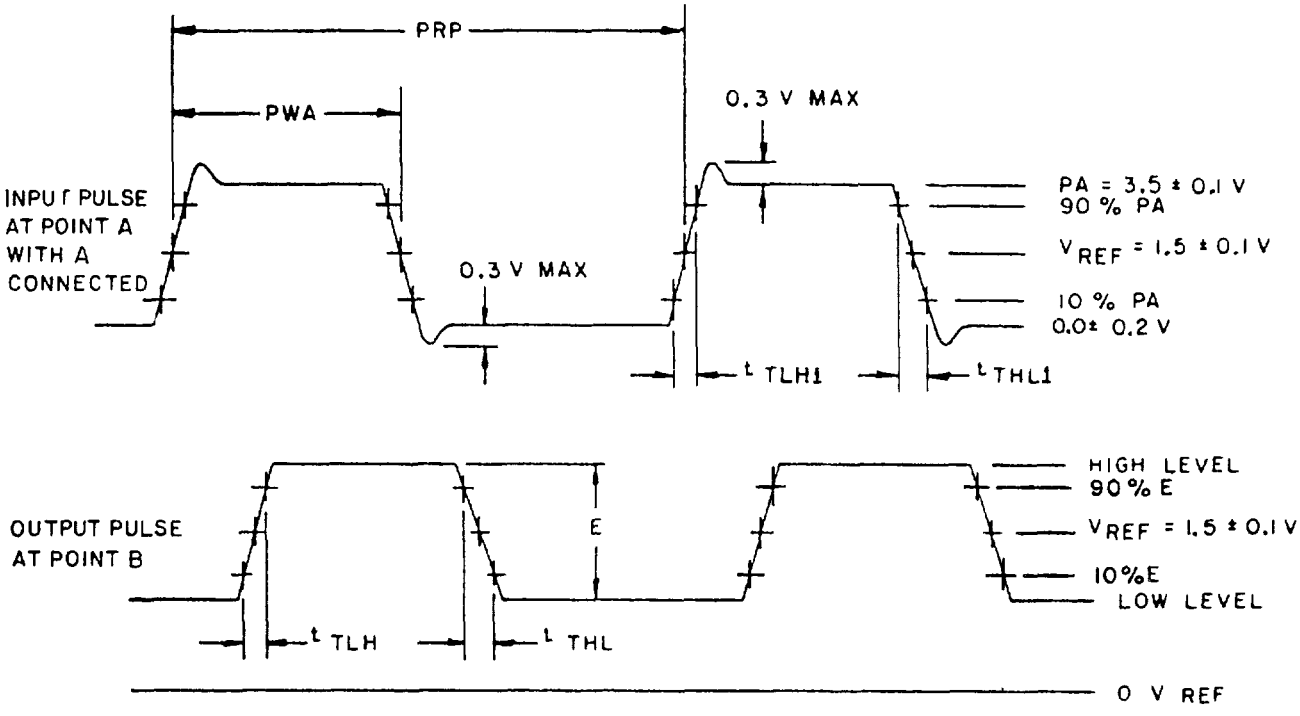


NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1.2 kΩ ±5%, 1/8 W.
3. S1 = SPST switch.
4. Load all outputs with the load as specified on figure 18.
5. Inititate memory loading procedure:

   a. Connect pins 2, 5, 32, 37, and 38 to pin 10, close switch S1 for a minimum of 500 ns, and then open switch. Memory location zero is then loaded with a low.
   b. Connect pins 2, 5, 32, 37, and 38 to pin 1, close switch S1 for a minimum of 500 ns, and then open switch. Memory location one is then loaded with a high.
   c. Connect point A as specified in table XV.

FIGURE 24. Transition time (t_{TLH} and t_{THL}) test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1,000 ±100 ns.
   PRP = 2,000 ±200 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.40 V.
3. All input voltages are referenced to ground.

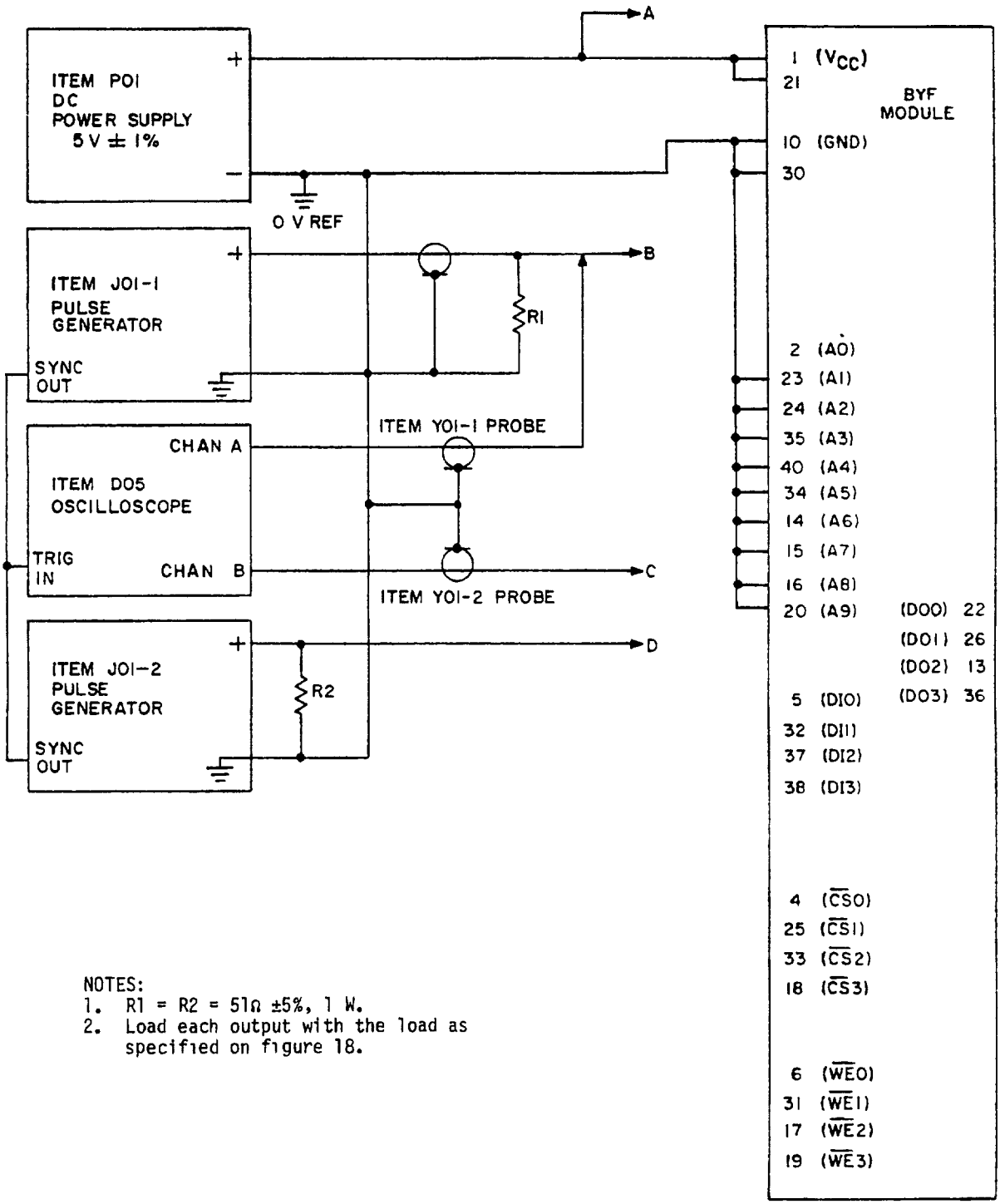
FIGURE 25. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XV.

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TABLE XVI. Select recovery time (t_{SRL} and t_{SRH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	$\overline{CS0}$ Select recovery time (t_{SRL})	6,17,18,19, 25,31,33	4	22	2	5	40	ns	26,27
R2	$\overline{CS1}$ "	4,6,17,18, 19,31,33	25	26	"	"	"	"	"
R3	$\overline{CS2}$ "	4,6,17,18, 19,25,31	33	13	"	"	"	"	"
R4	$\overline{CS3}$ "	4,6,17,19, 25,31,33	18	36	"	"	"	"	"
R5	$\overline{CS0}$ Select recovery time (t_{SRH})	6,17,18,19, 25,31,33	4	22	"	"	"	"	"
R6	$\overline{CS1}$ "	4,6,17,18, 19,31,33	25	26	"	"	"	"	"
R7	$\overline{CS2}$ "	4,6,17,18, 19,25,31	33	13	"	"	"	"	"
R8	$\overline{CS3}$ "	4,6,17,19, 25,31,33	18	36	"	"	"	"	"

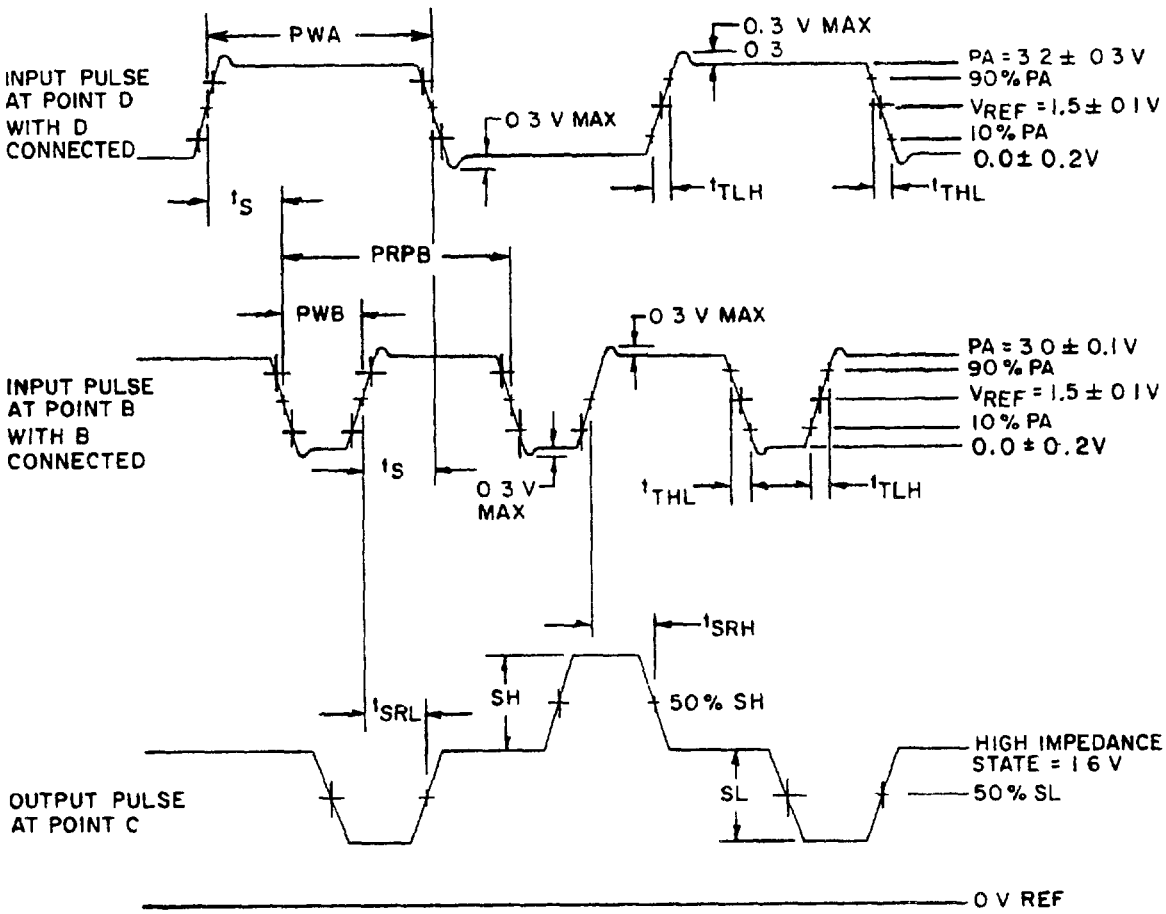
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load each output with the load as specified on figure 18.

FIGURE 26. Select recovery time (t_{SRL} and t_{SRH}) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PWA = 2,000 ±100 ns.
   PWB = 600 ±50 ns.
   PRPB= 2,000 ±100 ns.
   t_S = 700 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Initial setup procedure.

   a. Store a high level in address location zero by connecting pins 2, 4, 18, 25, and 33 to pin 10 and connecting pins 5, 32, 37, and 38 to pin 1. Connect pins 6, 17, 19, and 31 to pin 10 for 400 ns minimum, then connect to point A.
   b. Store a low level in address location one by connecting pin 2 to pin 1 and connecting pins 4, 5, 18, 25, 32, 33, 37, and 38 to pin 10. Connect pins 6, 17, 19, and 31 to pin 10 for 400 ns minimum, then connect to point A.
   c. Verify that address location zero has a high stored and location one has a low stored.

FIGURE 27. Select recovery time (t_{SRL} and t_{SRH}) test circuit waveforms for table XVI.

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TABLE XVII. Dynamic test.

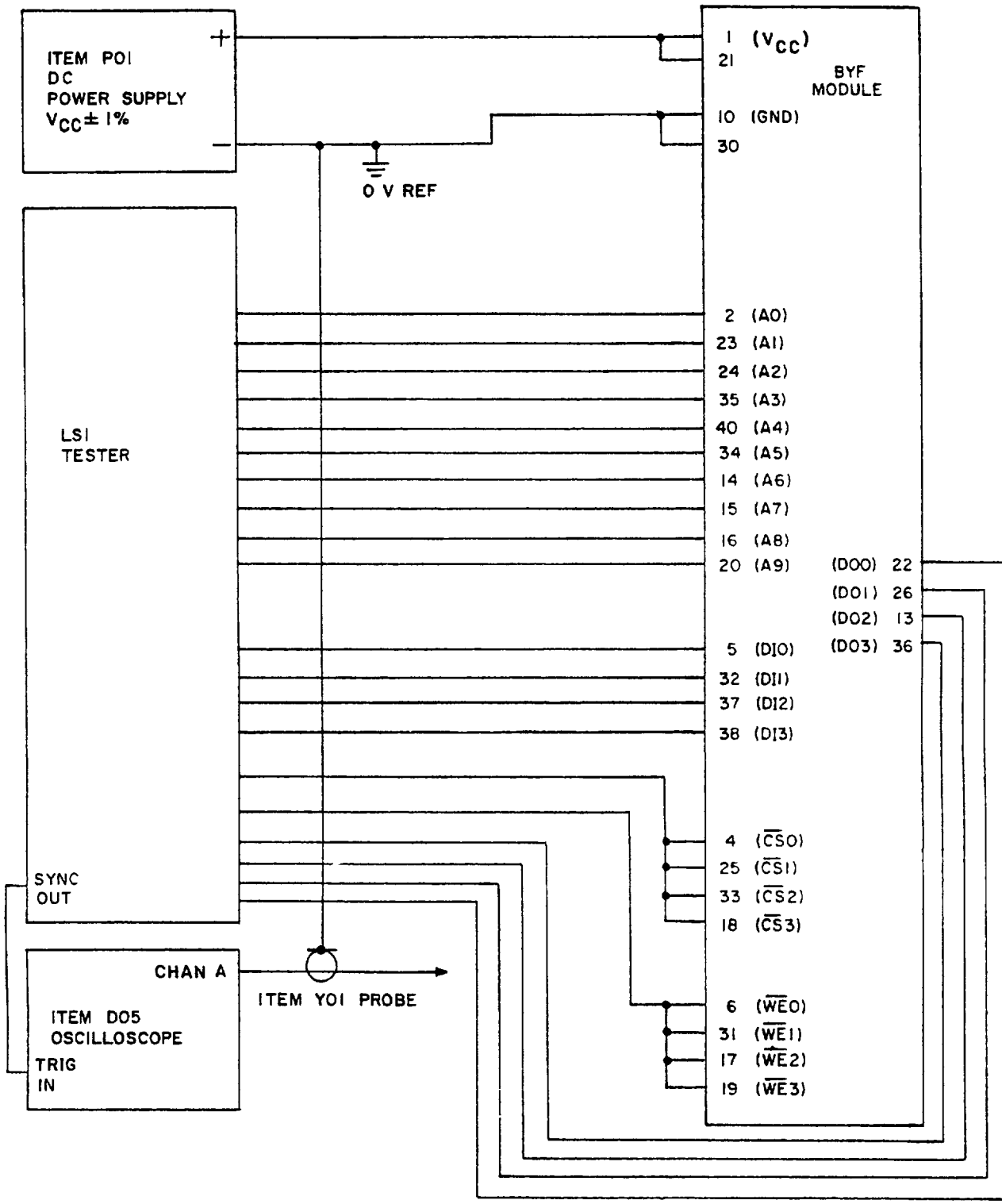
Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S1	Galloping-one, galloping-zero test pattern, C_L = 50 pF, V_{CC} = 4.5 V	Module shall conform to figure 30 requirements			28,29,30
S2	Galloping-one, galloping-zero test pattern, C_L = 50 pF, V_{CC} = 5.5 V	Module shall conform to figure 30 requirements			28,29,30
S3	Inter-address write-recovery-one, inter-address write-recovery-zero test pattern, C_L = 50 pF, V_{CC} = 4.5 V	Module shall conform to figure 31 requirements			28,29,31
S4	Inter-address write-recovery-one, inter-address write-recovery-zero test pattern, C_L = 50 pF, V_{CC} = 5.5 V	Module shall conform to figure 31 requirements			28,29,31
S5	Marching-write-recovery-one, marching-write-recovery-zero and circuit select/deselect test pattern, C_L = 50 pF, V_{CC} = 4.5 V	Module shall conform to figure 32 requirements			28,29,32
S6	Marching-write-recovery-one, marching-write-recovery-zero and circuit select/deselect test pattern, C_L = 50 pF, V_{CC} = 5.5 V	Module shall conform to figure 32 requirements			28,29,32
S7	Static hold test pattern, C_L = 50 pF, V_{CC} = 4.5 V	Module shall conform to figure 33 requirements			28,29,33
S8	Static hold test pattern, C_L = 50 pF, V_{CC} = 5.5 V	Module shall conform to figure 33 requirements			28,29,33
S9	Galloping-one, galloping-zero test pattern, C_L = 150 pF, V_{CC} = 4.5 V	Module shall conform to figure 30 requirements			28,29,30
S10	Galloping-one, galloping-zero test pattern, C_L = 150 pF, V_{CC} = 5.5 V	Module shall conform to figure 30 requirements			28,29,30
S11	Inter-address write-recovery-one, inter-address write-recovery-zero test pattern, C_L = 150 pF, V_{CC} = 4.5 V	Module shall conform to figure 31 requirements			28,29,31
S12	Inter-address write-recovery-one, inter-address write-recovery-zero test pattern, C_L = 150 pF, V_{CC} = 5.5 V	Module shall conform to figure 31 requirements			28,29,31

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TABLE XVII. Dynamic test - Continued.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S13	Marching-write-recovery-one, marching-write-recovery-zero and circuit select/deselect test pattern, C_L = 150 pF, V_{CC} = 4.5 V	Module shall conform to figure 32 requirements			28,29,32
S14	Marching-write-recovery-one, marching-write-recovery-zero and circuit select/deselect test pattern, C_L = 150 pF, V_{CC} = 5.5 V	Module shall conform to figure 32 requirements			28,29,32
S15	Static hold test pattern, C_L = 150 pF, V_{CC} = 4.5 V	Module shall conform to figure 33 requirements			28,29,33
S16	Static hold test pattern, C_L = 150 pF, V_{CC} = 5.5 V	Module shall conform to figure 33 requirements			28,29,33

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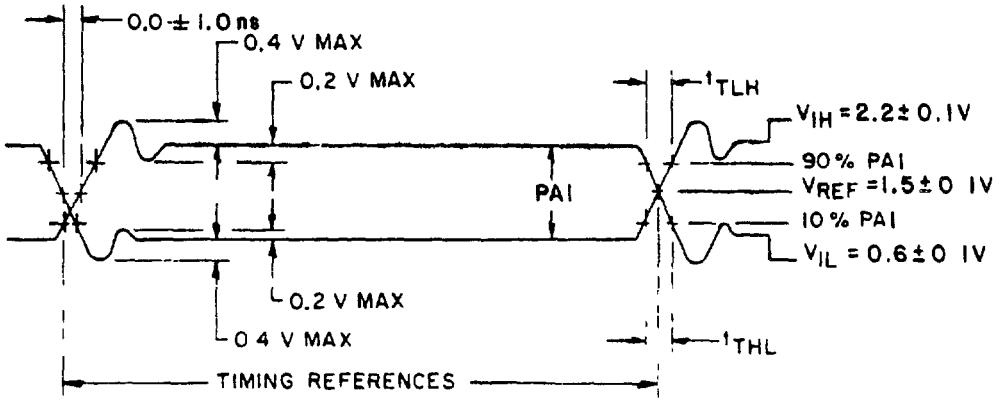


NOTES:
1. V_{CC} = 4.5 V ±1% for test conditions S1, S3, S5, S7, S9, S11, S13 and S15.
2. V_{CC} = 5.5 V ±1% for test conditions S2, S4, S6, S8, S10, S12, S14, and S16.
3. Load output pins 13, 22, 26, and 36 with the load as specified on figure 18.

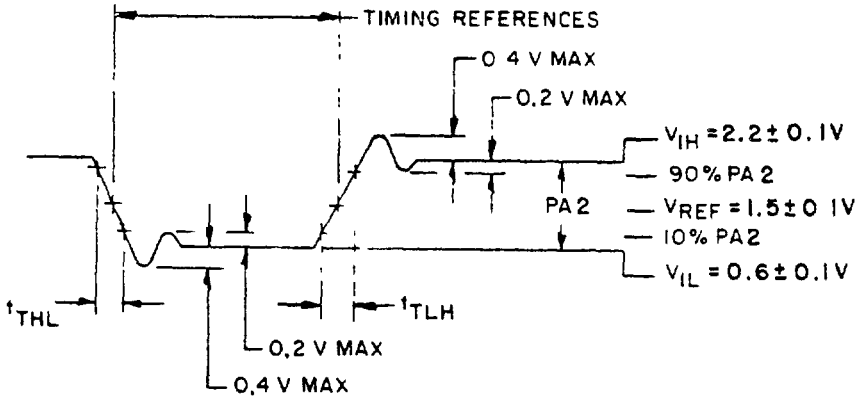
FIGURE 28. Dynamic test circuit for table XVII.

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EDGE SKEW BIDIRECTIONAL INPUT WAVEFORMS



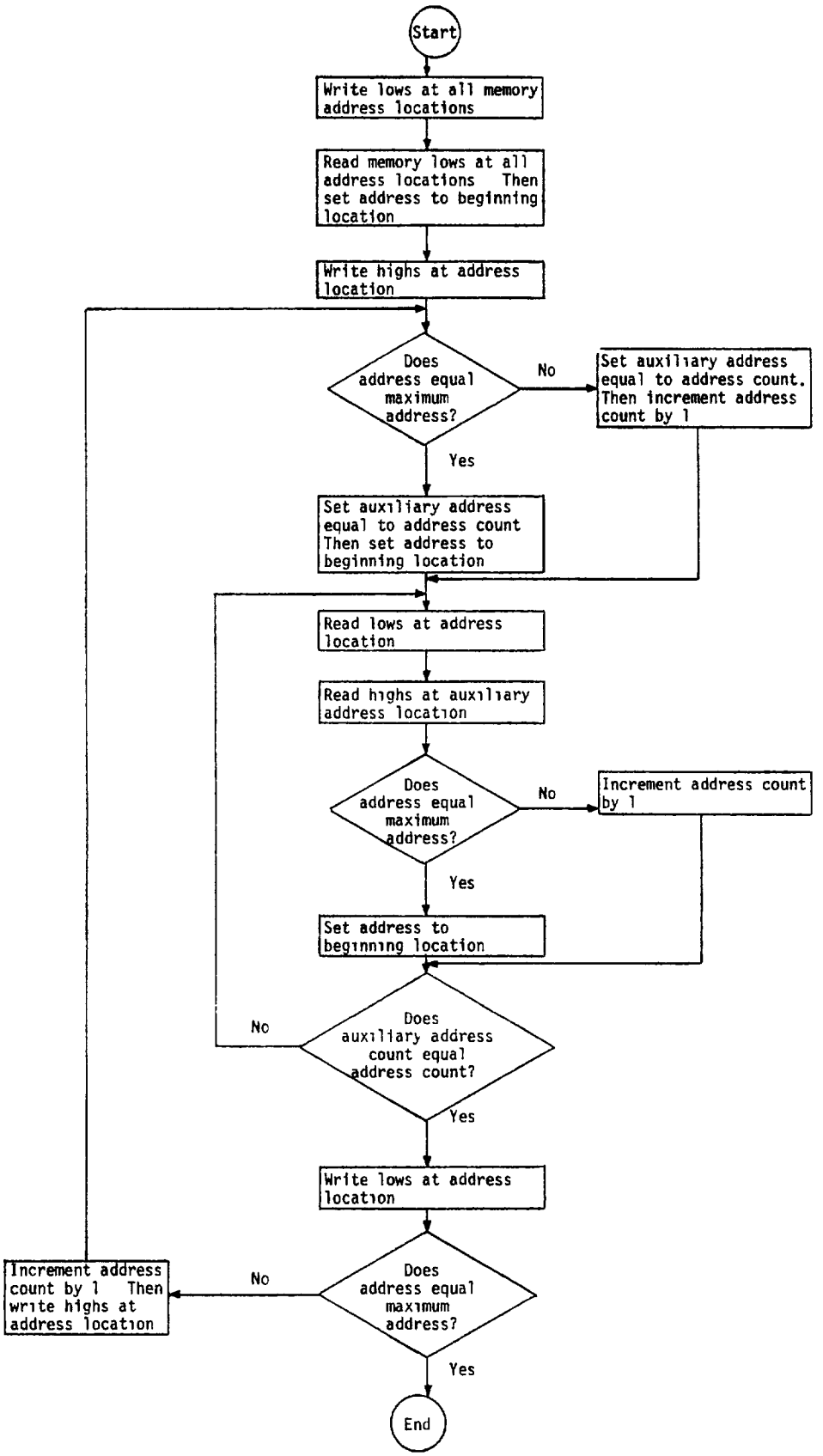
UNIDIRECTIONAL INPUT WAVEFORMS



NOTES.
1. $t_{THL} = t_{TLH} = 10 \pm 4$ ns.
2. All timing references are made at the V_{REF} level.

FIGURE 29. Dynamic test circuit waveforms for table XVII.

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NOTE. This flow chart applies to one test type only and may be used for the other test as indicated by interchanging "highs" and "lows"

FIGURE 30 Galloping-one, galloping-zero test pattern program flow chart, test waveforms, and program for table XVII.

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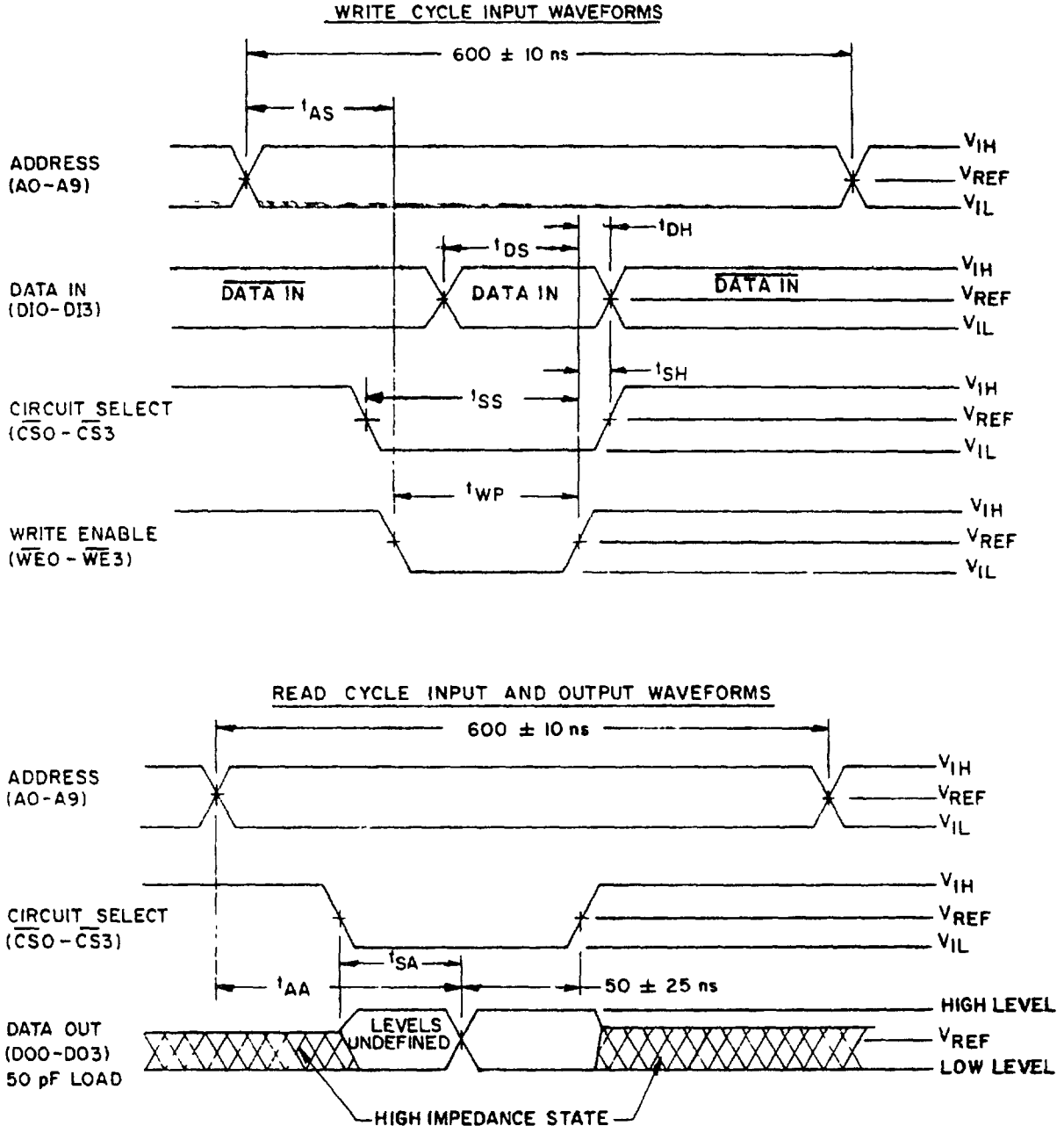


FIGURE 30. Galloping-one, galloping-zero test pattern program flow chart, test waveforms, and program for table XVII - Continued.

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NOTES:

1. The objective of galloping-one-galloping-zero test pattern is to test access time. It is necessary to check all possible address transitions (addressing) from a read cycle to a read cycle. This pattern also verifies minimum write pulse width (t_{WP}), address setup time (t_{AS}), circuit-select setup time (t_{SS}), circuit-select hold time (t_{SH}), data setup time (t_{DS}), data hold time (t_{DH}), circuit select access time (t_{SA}), and address access time (t_{AA}).

2. A program that fulfills the galloping-one test pattern requirements would:

   a. Write "lows" in all memory address locations.
   b. Read "lows" in all memory address locations.
   c. Write a "high' in memory address location zero.
   d. Read a "low" in memory address location one.
   e. Read a "high" in memory address location zero.
   f. Read a "low" in memory address location two.
   g. Read a "high" in memory address location zero.
   h. Read a "low" in memory address location three.
   i. Read a "high" in memory address location zero.
   j. Read a "low" in memory address location four.
   k. Continue this pattern until memory address location zero has been checked with respect to all other memory address locations.
   l. Write a "low" in memory address location zero.
   m. Write a "high" in memory address location one.
   n. Read a "low" in memory address location two.
   o. Read a "high" in memory address location one.
   p. Read a "low" in memory address location three.
   q. Read a "high" in memory address location one.

      .
      .
      .

   r. Read a "low" in memory address location 1023.
   s. Read a "high" in memory address location one.
   t. Read a "low" in memory address location zero.
   u. Read a "high" in memory address location one
   v. Continue the pattern until a "high" has been moved through all memory address locations.

3. A program that fulfills the galloping-zero test pattern requirements would be as in note 2, and figure 30 flow chart, with all "highs" and "lows" interchanged.

4. Pattern test waveform parameters:

   a. Address setup time (t_{AS}) = 150 +0, -5 ns.
   b. Circuit select setup time (t_{SS}) = 300 +0, -5 ns.
   c. Circuit select hold time (t_{SH}) = 50 +0, -5 ns.
   d. Data setup time (t_{DS}) = 300 +0, -5 ns.

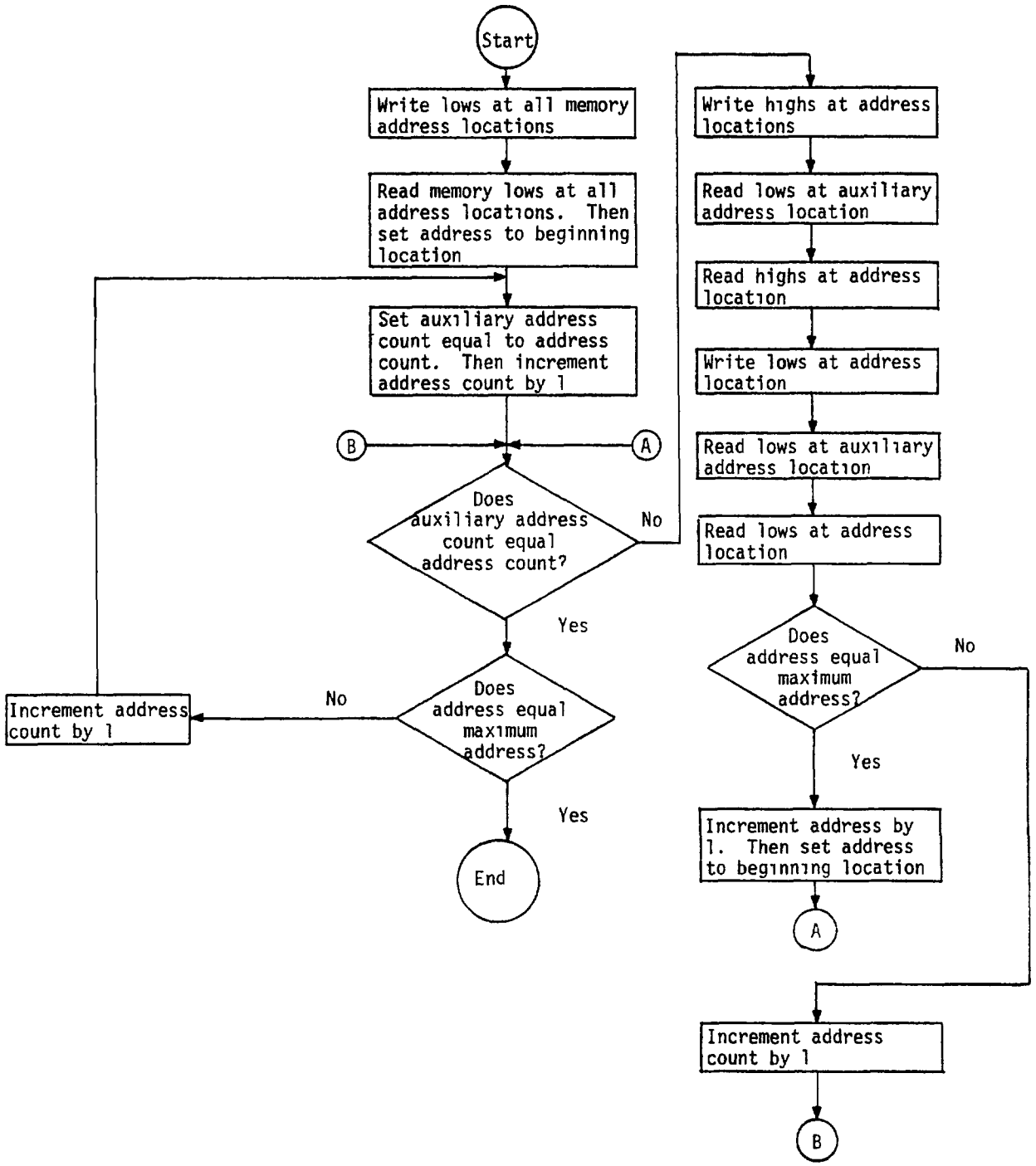
FIGURE 30. Galloping-one, galloping-zero test pattern program flow chart, test waveforms, and program for table XVII - Continued.

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e. Data hold time (t_{DH}) = 100 +0, -5 ns.
f. Write pulse width (t_{WP}) = 300 +0, -5 ns.
g. Address access time, 50 pF load (t_{AA}) = 525 +0, -5 ns.
h. Address access time, 150 pF load (t_{AA}) = 530 +0, -5 ns.
i. Circuit select access time, 50 pF load (t_{SA}) = 375 +0, -5 ns.
j. Circuit select access time, 150 pF load (t_{SA}) = 380 +0, -5 ns.
k. V_{IH} = 2.2 ±0.1 V.
l. V_{IL} = 0.6 ±0.1 V.
m. V_{REF} = 1.5 ±0.1 V.
n. All waveforms shall conform to the detailed requirements of figure 29.
o. All address lines need not switch simultaneously each cycle.

FIGURE 30. Galloping-one, galloping-zero test pattern program flow chart, test waveforms, and program for table XVII - Continued.

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NOTE: This flow chart applies to one test type only and may be used for the other test as indicated by interchanging highs and lows.

FIGURE 31 Inter-address write-recovery-one, inter-address write-recovery-zero test pattern program flow chart, test waveforms, and program for table XVII.

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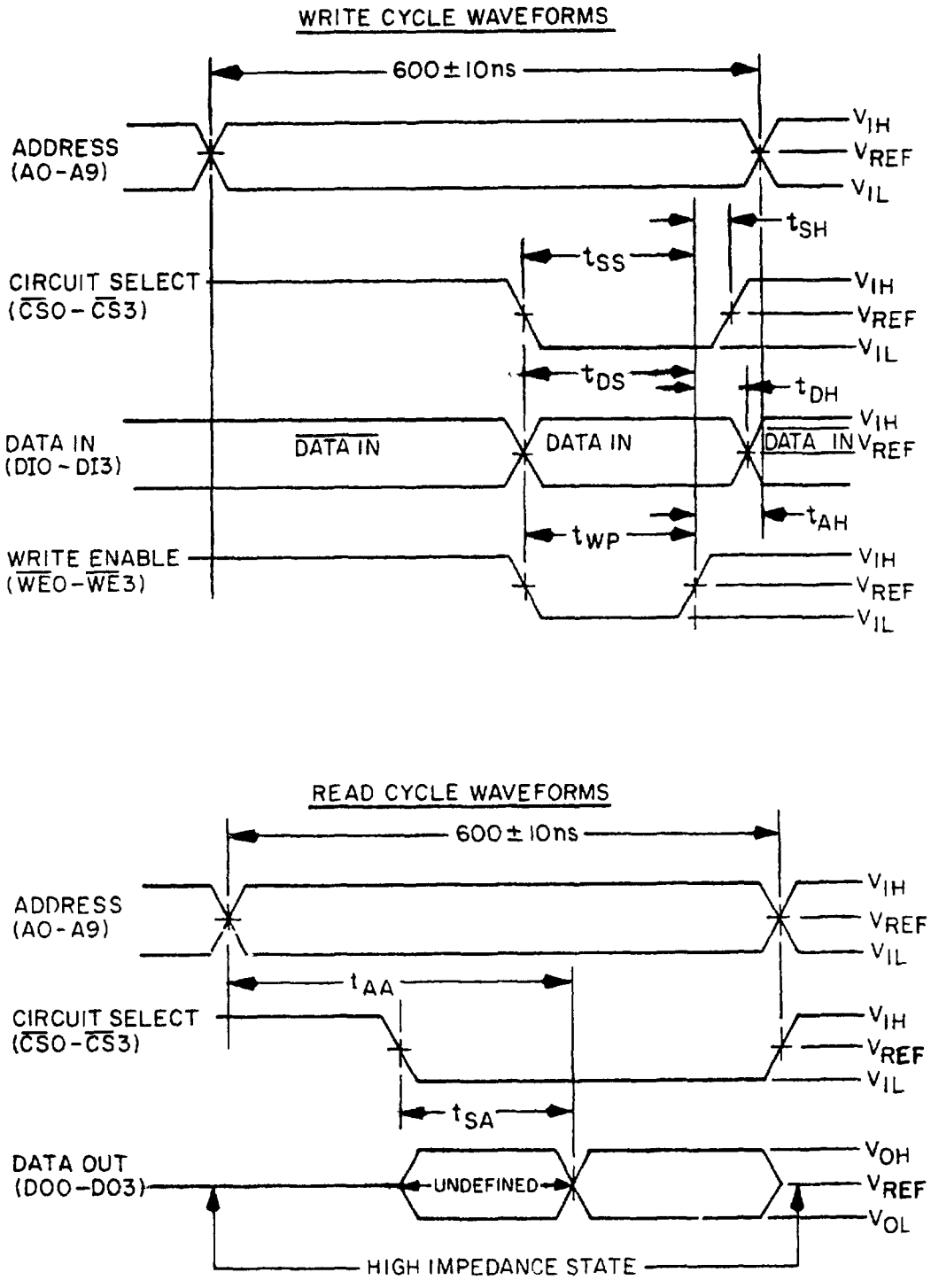


FIGURE 31. Inter-address write-recovery-one, inter-address write-recovery-zero test pattern program flow chart, test waveforms, and program for table XVII - Continued.

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NOTES:

1. The objective of inter-address write-recovery-one, inter-address write-recovery-zero test pattern is to test the recovery time from a write at one address location to a read at a different address location. It is necessary to check all possible address transitions from a write cycle at one address to a read cycle at a different address. This pattern also verifies address hold time (t_{AH}), minimum write pulse width (t_{WP}), address access time (t_{AA}), circuit select access time (t_{SA}), data setup time (t_{DS}), circuit select setup time (t_{SS}) and data hold time (t_{DH}).

2. A program that fulfills the inter-address write-recovery-one test pattern requirements would:

   a. Write "lows" at all memory address locations.
   b. Read "lows" at all memory address locations.
   c. Write "highs" at memory address location one.
   d. Read "lows" at memory address location zero.
   e. Read "highs" at memory address location one.
   f. Write "lows" at memory address location one.
   g. Read "lows" at memory address location zero.
   h. Read "lows" at memory address location one.
   i. Continue the pattern until memory address location zero has been checked with respect to all other memory address locations.
   j. Write "highs" at memory address location two.
   k. Read "lows" at memory address location one.
   l. Read "highs" at memory address location two.
   m. Write "lows" at memory address location two.
   n. read "lows" at memory address location one.
   o. Read "lows" at memory address location two.

      .
      .
      .

   p. Write "highs" at memory address location zero.
   q. Read "lows" at memory address location one.
   r. Read "highs" at memory address location zero.
   s. Write "lows" at memory address location zero.
   t. Read "lows" at memory address location one.
   u. Read "lows" at memory address location zero.
   v. Continue the pattern until all memory address locations have been checked with respect to all other memory address locations.

3. A program that fulfills the inter-address write-recovery-zero test pattern requirements would be as in note 2, and figure 31 flow chart, with all "highs" and "lows" interchanged.

4. Pattern test waveform parameters:

   a. Address hold time (t_{AH}) = 100 +2, -5 ns.
   b. Write pulse width (t_{WP}) = 300 +0, -5 ns.
   c. Circuit select access time, 50 pF load (t_{SA}) = 375 +0, -5 ns.
   d. Circuit select access time, 150 pF load (t_{SA}) = 380 +0, -5 ns.
   e. Address access time, 50 pF load (t_{AA}) = 525 +0, -5 ns.
   f. Address access time, 150 pF load (t_{AA}) = 530 +0, -5 ns.
   g. Data hold time (t_{DH}) = 100 +0, -5 ns.

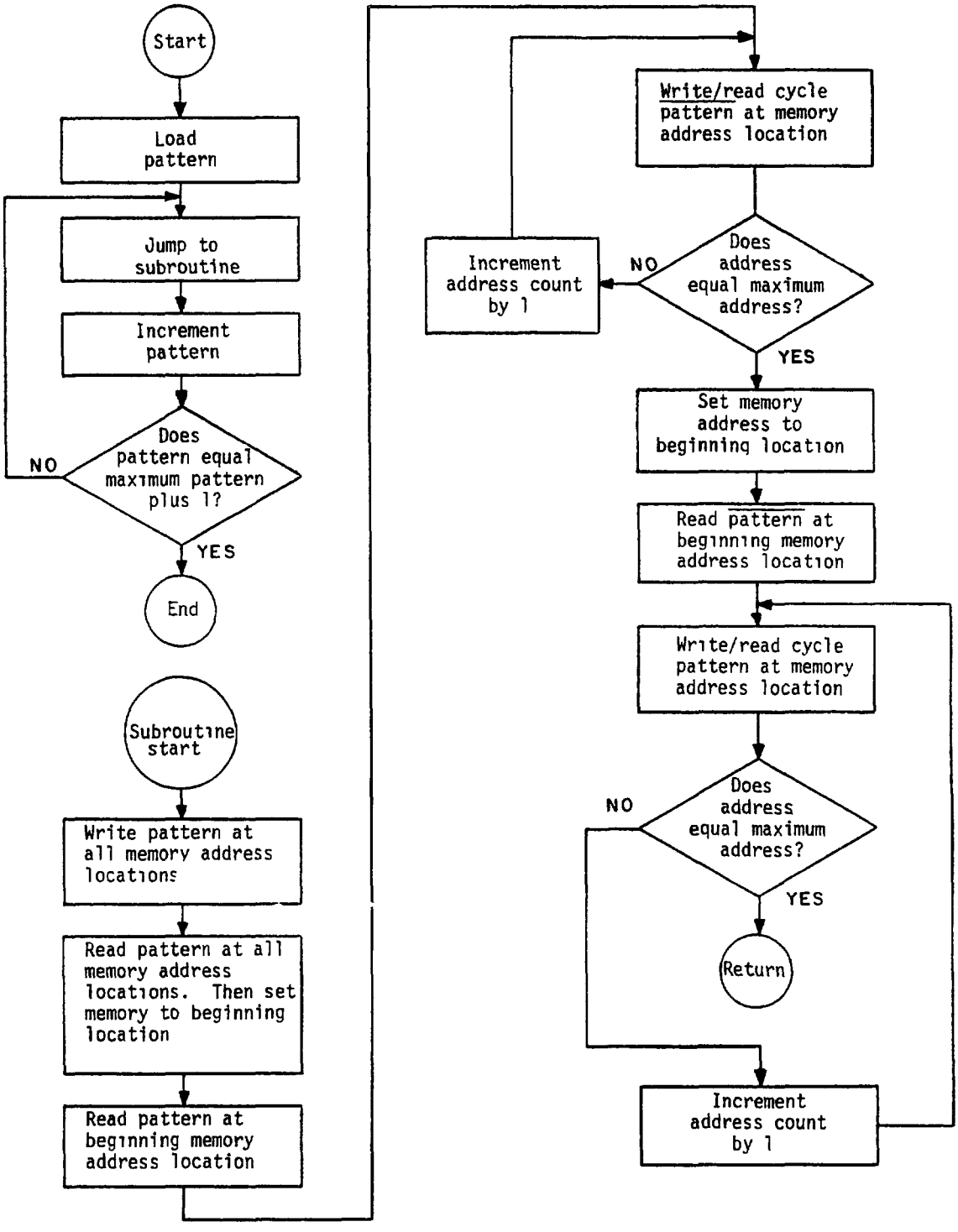
FIGURE 31. Inter-address write-recovery-one, inter-address write-recovery-zero test pattern program flow chart, test waveforms, and program for table XVII - Continued.

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h. Circuit select setup time (t_{SS}) = 300 +0, -5 ns.
i. Circuit select hold time (t_{SH}) = 50 +0, -5 ns.
j. Data setup time (t_{DS}) = 300 +0, -5 ns.
k. V_{IH} = 2.2 ±0.1 V.
l. V_{IL} + 0.6 ±0.1 V.
m. V_{REF} = 1.5 ±0.1 V.
n. All waveforms shall conform to the detailed input waveform requirements of figure 29.
o. All address lines need not switch simultaneously each cycle.

FIGURE 31. Inter-address write-recovery-one, inter-address write-recovery-zero test pattern program flow chart, test waveforms, and program for table XVII - Continued.

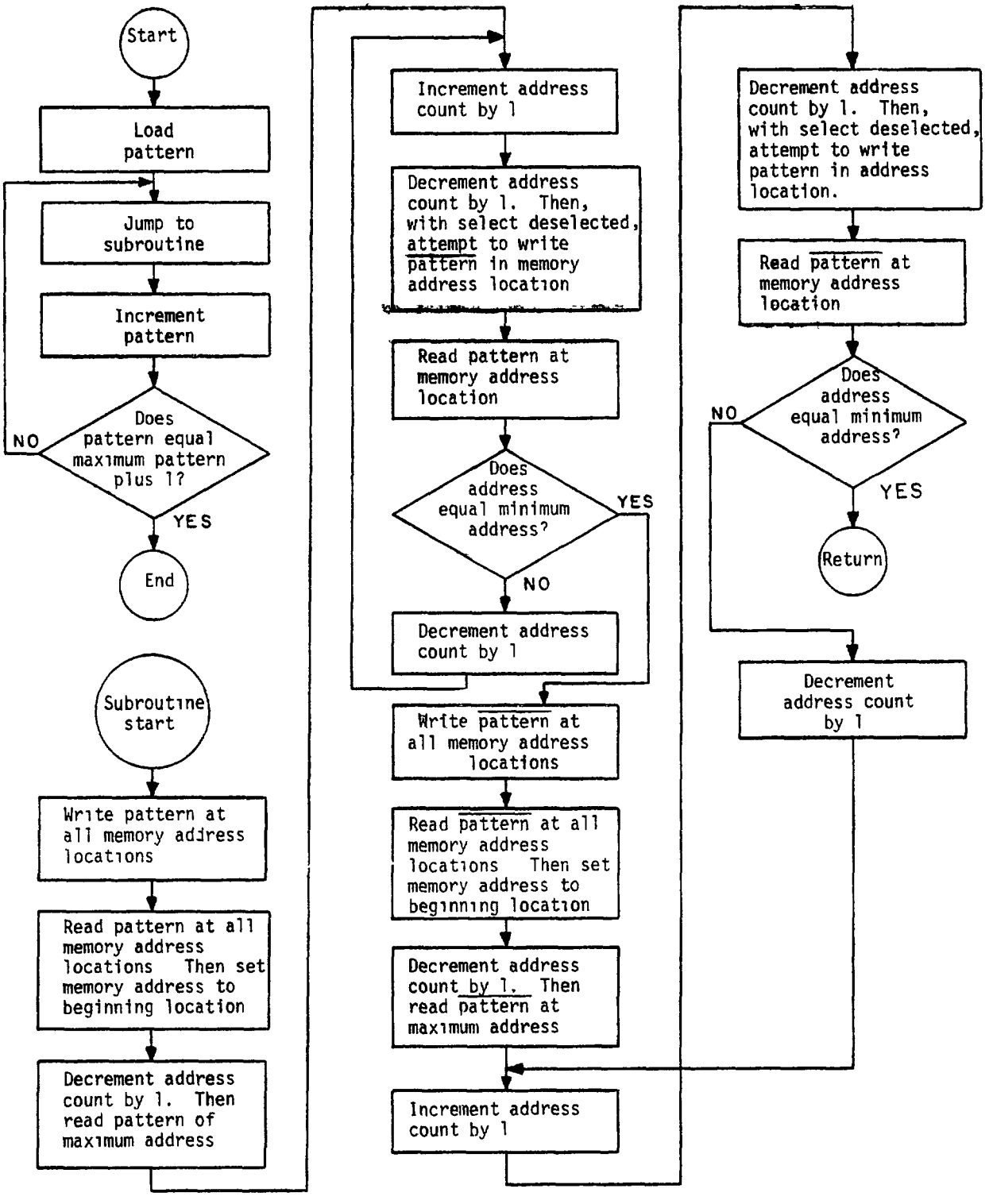
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MARCHING-WRITE-RECOVERY-ONE, MARCHING-WRITE-RECOVERY-ZERO PATTERN FLOW CHART

FIGURE 32. Marching-write-recovery-one, marching-write-recovery-zero test pattern flow chart, circuit select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII.

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CIRCUIT SELECT/DESELECT PATTERN FLOW CHART

NOTE: This pattern shall be repeated four times using the data pattern specified in note 2.

FIGURE 32. Marching-write-recovery-one, marching-write-recovery-zero test pattern flow chart, circuit select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII - Continued.

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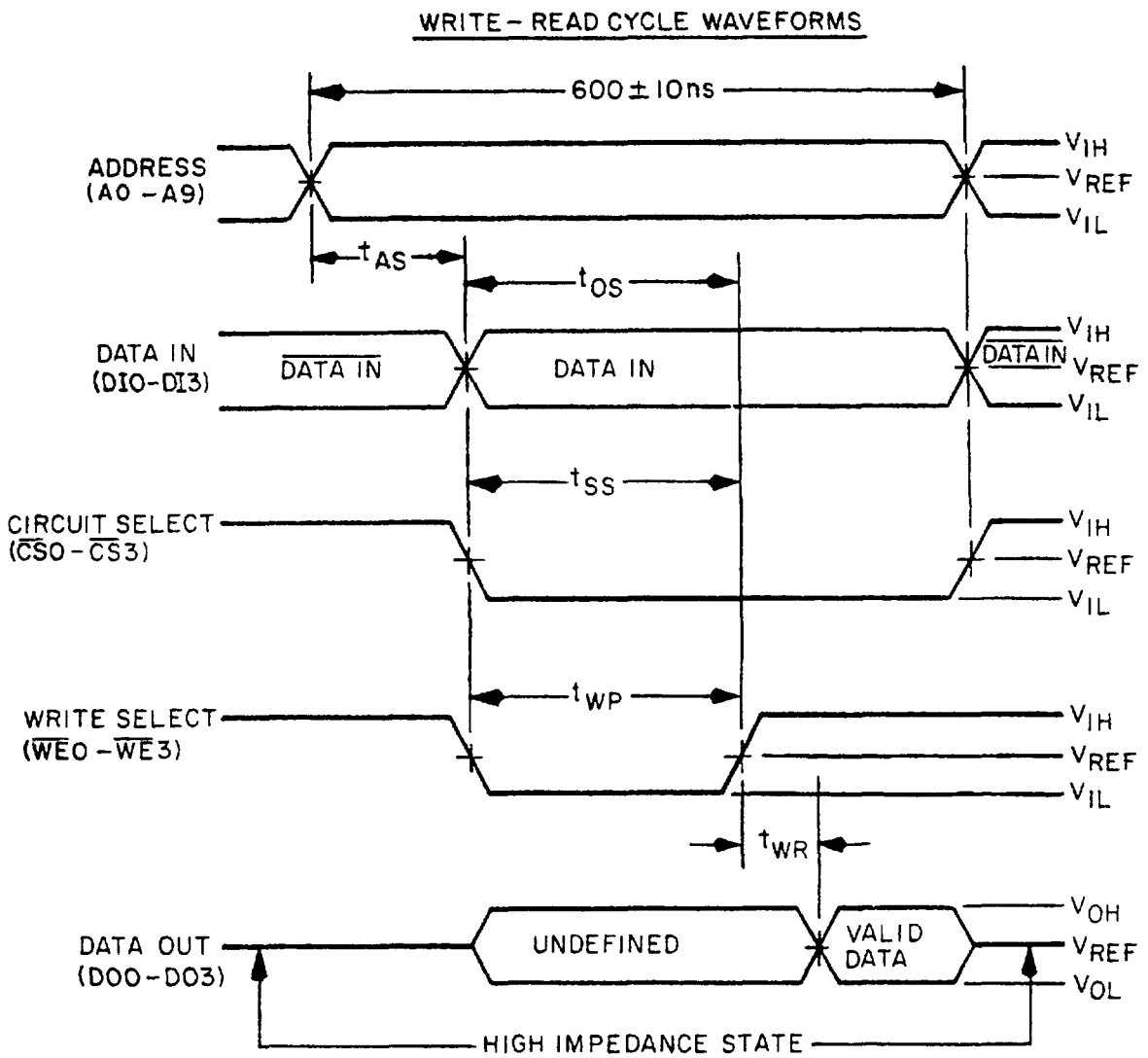


FIGURE 32. Marching-write-recovery-one, marching-write-recovery-zero test pattern flow chart, circuit select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII - Continued.

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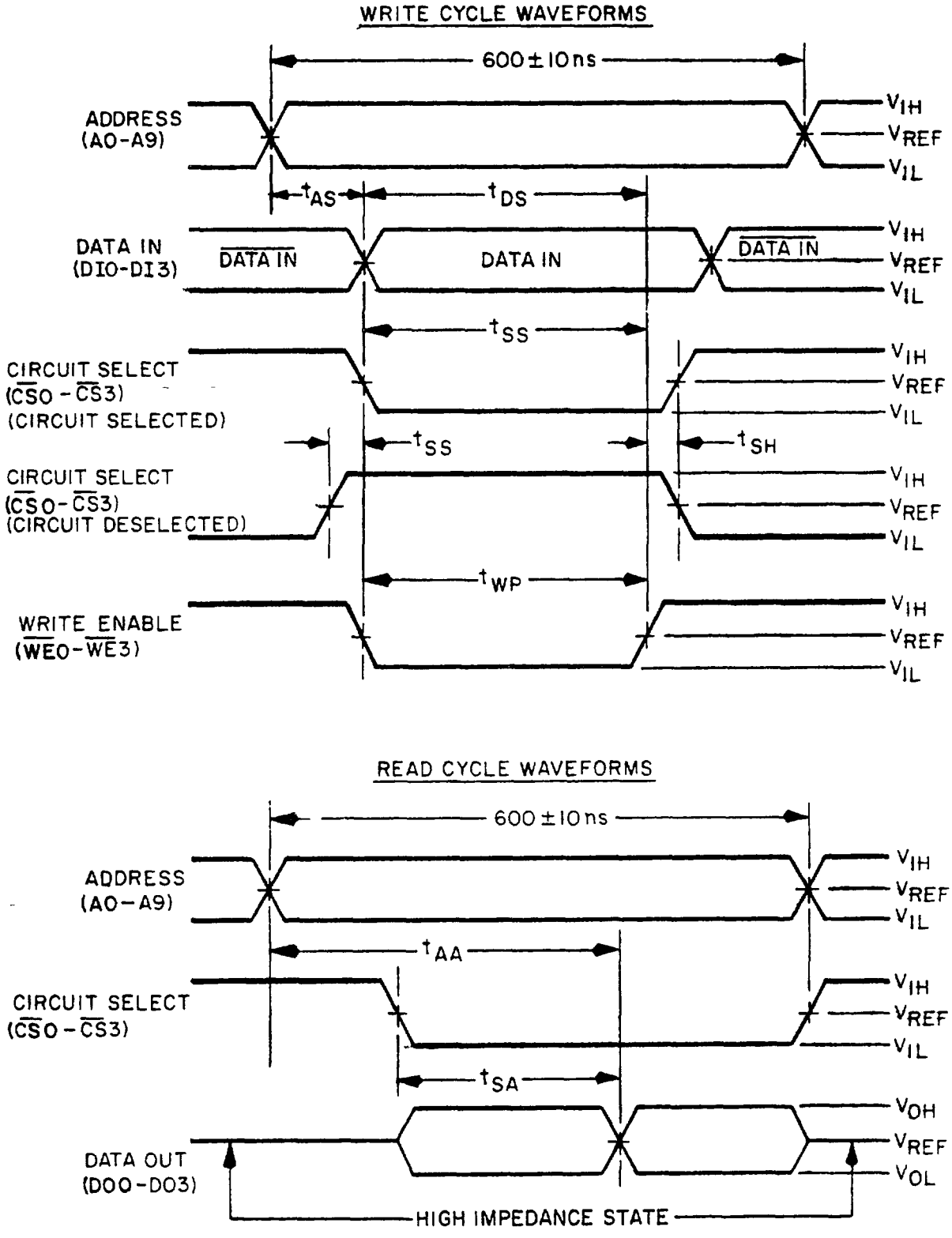


FIGURE 32. Marching-write-recovery-one, marching-write-recovery-zero test pattern flow chart, circuit select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII - Continued.

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NOTES:

1. The objective of the marching-write-recovery-one, marching-write-recovery-zero test pattern is to check write recovery time of the module at the same address. The objective of the circuit select/deselect test pattern is to check for proper functioning of the address and circuit select circuitry in a dynamic environment. The patterns also verify address setup time (t_{AS}), circuit select setup time ($t_{\overline{SS}}$), circuit deselect setup time (t_{SS}), circuit select hold time (t_{SH}), write recovery time (t_{WR}), write pulse width (t_{WP}), address access time (t_{AA}), select access time (t_{SA}), and data setup time (t_{DS}).

2. The chip independence test is a data pattern that is run during the marching-write-recovery test and the chip select/deselect test. This pattern checks that writing to one data bit does not affect other data bits. To run this pattern, the entire marching-write-recovery-one, marching-write-recovery-zero and chip select/deselect patterns are run on each of the following data patterns and their complements.

	DI3	DI2	DI1	DI0
A	0	0	0	1
B	0	0	1	0
C	0	1	0	0
D	1	0	0	0

3. A program that fulfills the marching-write-recovery-one, marching-write-recovery-zero test pattern requirements would:

   a. Write and read pattern A at all memory address locations.
   b. Read pattern A at address location zero.
   c. Write, then read pattern $\overline{A}$ at address location zero.
   d. Repeat (b) and (c) for address locations one through 1023, in turn.
   e. Set memory address location to zero.
   f. Read pattern $\overline{A}$ at address location zero.
   g. Write, then read a low at address location zero.
   h. Repeat (f) and (g) for address locations one through 1023, in turn.
   i. Repeat steps (a) through (h) with patterns B, C, and D.

4. A program that fulfills the circuit select/deselect test pattern requirements would:

   a. Write and read pattern A at all memory address locations.
   b. Set memory address to location zero.
   c. Decrement memory address by one, then read pattern A at maximum address.
   d. Increment memory address location by one.
   e. Decrement memory address location by one, then, with select deselected, attempt to write pattern $\overline{A}$ at the memory address location.
   f. Read pattern A at memory address location.
   g. Decrement address location by one and repeat (d) through (f) for the resultant address location.
   h. Repeat (g) until reading address location zero.
   i. Repeat steps (a) through (h) pattern A and pattern $\overline{A}$ interchanged.
   j. Repeat steps (a) through (i) with patterns B, C, and D.

FIGURE 32. Marching-write-recovery-one, marching-write-recovery-zero test pattern flow chart, circuit select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII - Continued.

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5. Pattern test waveform parameters:

a. Address setup time (t_{AS}) = 150 +0, -5 ns.
b. Circuit select setup time (t_{SS}) = 300 +0, -5 ns.
c. Circuit deselect time (t_{SS}) = 50 +3, -0 ns.
d. Circuit select hold time (t_{SH}) = 50 +3, -0 ns.
e. Data setup time (t_{DS}) = 300 +0, -5 ns.
f. Write pulse width (t_{WP}) = 300 +0, -5 ns.
g. Address access time, 50 pF load (t_{AA}) = 525 +0, -10 ns.
h. Address access time, 150 pF load (t_{AA}) = 530 +0, -10 ns.
i. Circuit select access time, 50 pF load (t_{SA}) = 375 +0, -5 ns.
j. Circuit select access time, 150 pF load (t_{SA}) = 380 +0, -5 ns.
k. Write recovery time, 50 pF load (t_{WR}) = 100 +0, -5 ns.
l. Write recovery time, 150 pF load (t_{WR}) = 110 +0, -5 ns.
m. V_{IH} = 2.2 ±0.1 V.
n. V_{IL} = 0.6 ±0.1 V.
o. V_{REF} = 1.5 ±0.1 V.
p. All waveforms shall conform to detailed input waveform requirements of figure 29.

FIGURE 32. Marching-write-recovery-one, marching-write-recovery-zero test pattern flow chart, circuit select/deselect and chip independence pattern program flow chart, test waveforms, and program for table XVII - Continued.

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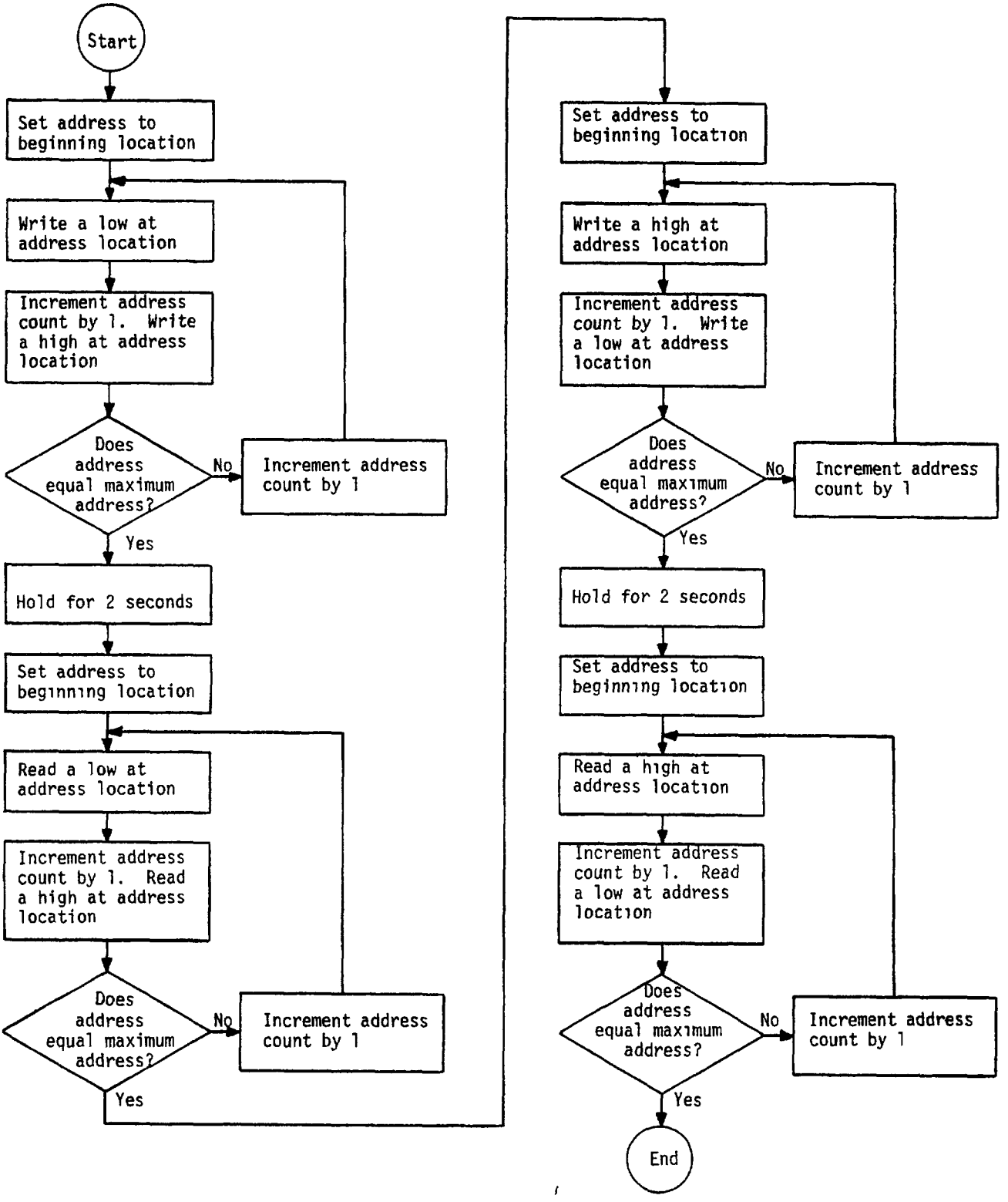


FIGURE 33. Static hold test pattern flow chart, test waveforms, and program for table XVII.

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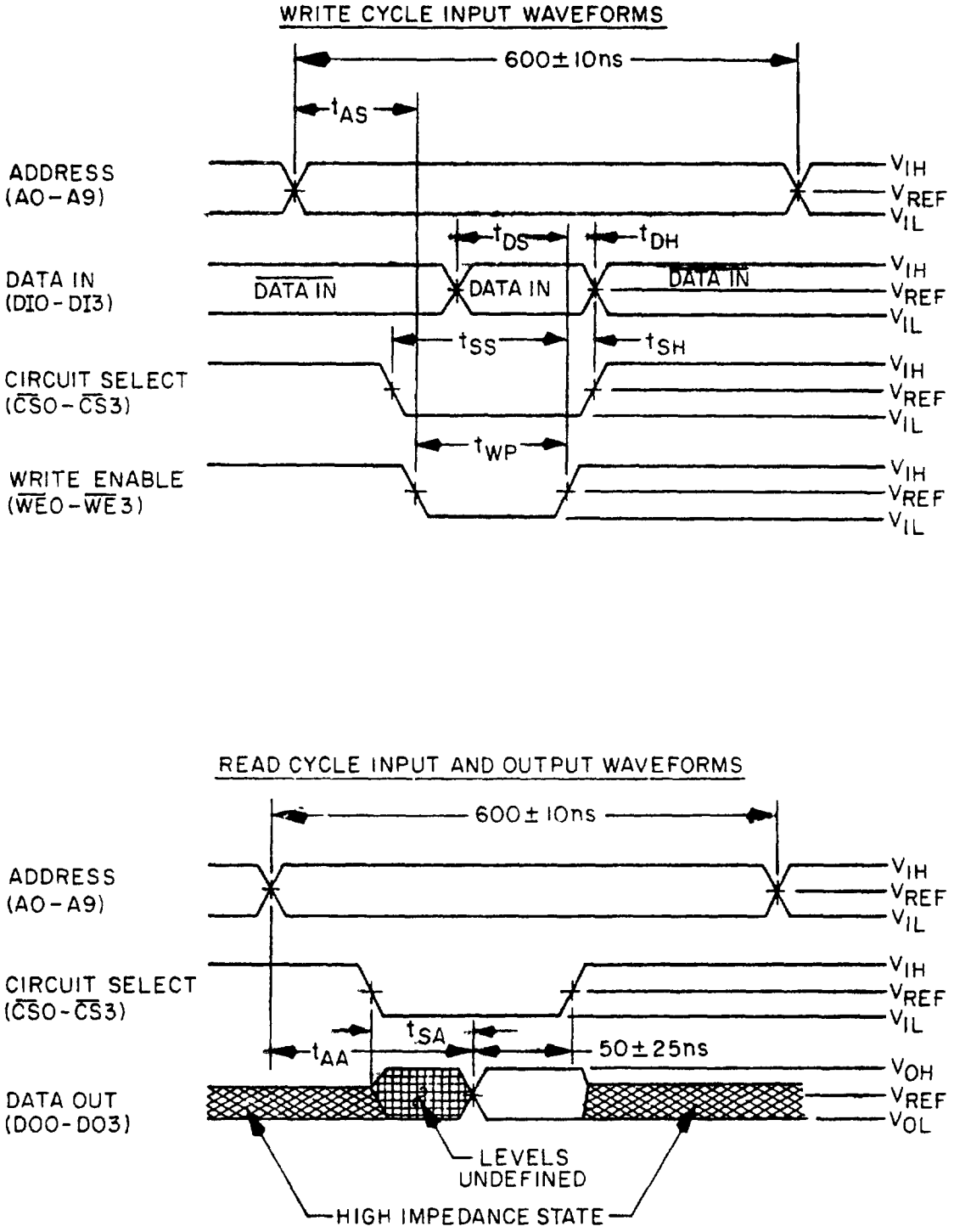


FIGURE 33. Static hold test pattern flow chart, test waveforms, and program for table XVII - Continued.

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NOTES:

1. The objective of the static hold test pattern is to check data retention after a static hold period of 2 seconds. The pattern also verifies minimum write pulse width (t_{WP}), address setup time (t_{AS}), select setup time (t_{SS}), data setup tim e(t_{DS}), select hold time (t_{SH}), data hold time (t_{DH}), select access time (t_{SA}), and address access time (t_{AA}).

2. A pattern that fulfills the static hold test pattern requirements would.

   a. Write a low at address location zero.
   b. Write a high at address location one.
   c. Repeat (a) and (b) for address locations two through 1023.
   d. Hold the memory (no transitions) for 2 seconds.
   e. Read a low at address location zero.
   f. Read a high at address location one.
   g. Repeat (e) and (f) for address locations two through 1023.
   h. Write a high at address locaton zero.
   i. Write a low at address location one.
   j. Repeat (h) and (i) for address locations two through 1023.
   k. Hold the memory (no transitions) for 2 seconds.
   l. Read a high at address location zero.
   m. Read a low at address location one.
   n. Repeat (l) and (m) for address locations two through 1023.

3. Pattern test waveform parameters:

   a. Address setup time (t_{AS}) = 150 +0, -5 ns.
   b. Data setup time (t_{DS}) = 300 +0, -5 ns.
   c. Setup hold time (t_{SS}) = 300 +0, -5 ns.
   d. Data hold time (t_{DH}) = 100 +0, -5 ns.
   e. Select hold time (t_{SH}) = 50 +0, -5 ns.
   f. Write pulse width (t_{WP}) = 300 +0, -5 ns.
   g. Address access time, 50 pF load (t_{AA}) = 525 +0, -5 ns.
   h. Address access time, 150 pF load (t_{AA}) = 530 +0, -5 ns.
   i. Select access time, 50 pF load (t_{SA}) = 375 +0, -5 ns.
   j. Select access time, 150 pF load (t_{SA}) = 380 +0, -5 ns.
   k. V_{IH} = 2.2 ±0.1 V.
   l. V_{IL} = 0.6 ±0.1 V.
   m. V_{REF} = 1.5 ±0.1 V.
   o. All waveforms shall conform to the detailed input waveform requirements of figure 29.
   p. All address lines need not switch simultaneously each cycle.

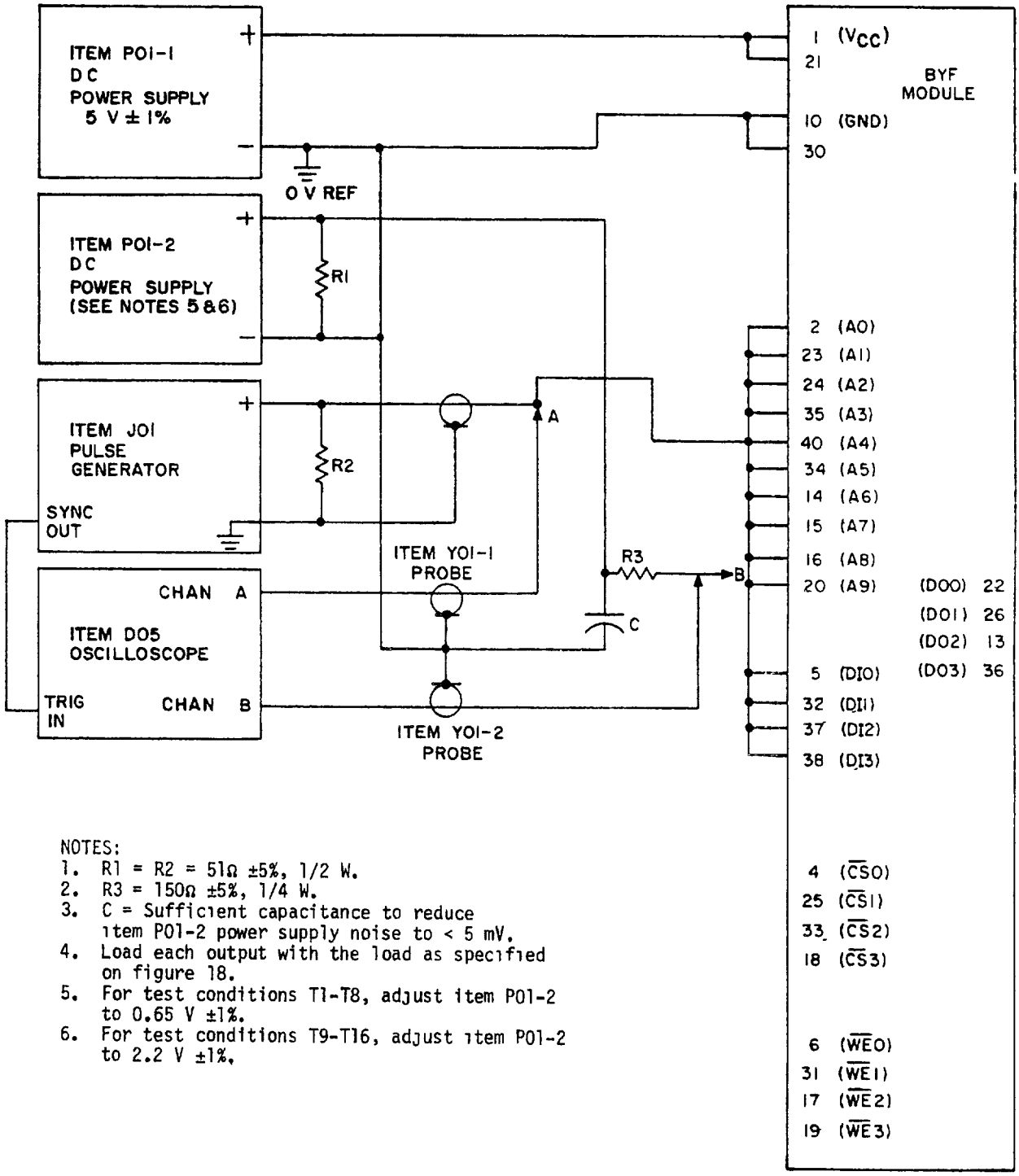
FIGURE 33. Static hold test pattern flow chart, test waveforms, and program for table XVII - Continued.

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Input crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	$\overline{CS0}$ Input crosstalk, low level	6,17,18,19,25, 31,33	4	0	900	mV	34,35
T2	$\overline{CS1}$ "	4,6,17,18,19, 31,33	25	"	"	"	"
T3	$\overline{CS2}$ "	4,6,17,18,19, 25,31	33	"	"	"	"
T4	$\overline{CS3}$ "	4,6,17,19,25, 31,33	18	"	"	"	"
T5	$\overline{WE0}$ "	4,17,18,19,25, 31,33	6	"	700	"	"
T6	$\overline{WE1}$ "	4,6,17,18,19, 25,33	31	"	"	"	"
T7	$\overline{WE2}$ "	4,6,18,19,25, 31,33	17	"	"	"	"
T8	$\overline{WE3}$ "	4,6,17,18,25, 31,33	19	"	"	"	"
T9	$\overline{CS0}$ Input crosstalk, high level	6,17,18,19,25, 31,33	4	"	"	"	"
T10	$\overline{CS1}$ "	4,6,17,18,19, 31,33	25	"	"	"	"
T11	$\overline{CS2}$ "	4,6,17,18,19, 25,31	33	"	"	"	"
T12	$\overline{CS3}$ "	4,6,17,19,25, 31,33	18	"	"	"	"
T13	$\overline{WE0}$ "	4,17,18,19,25, 31,33	6	"	500	"	"
T14	$\overline{WE1}$ "	4,6,17,18,19, 25,33	31	"	"	"	"
T15	$\overline{WE2}$ "	4,6,18,19,25, 31,33	17	"	"	"	"
T16	$\overline{WE3}$ "	4,6,17,18,25, 31,33	19	"	"	"	"

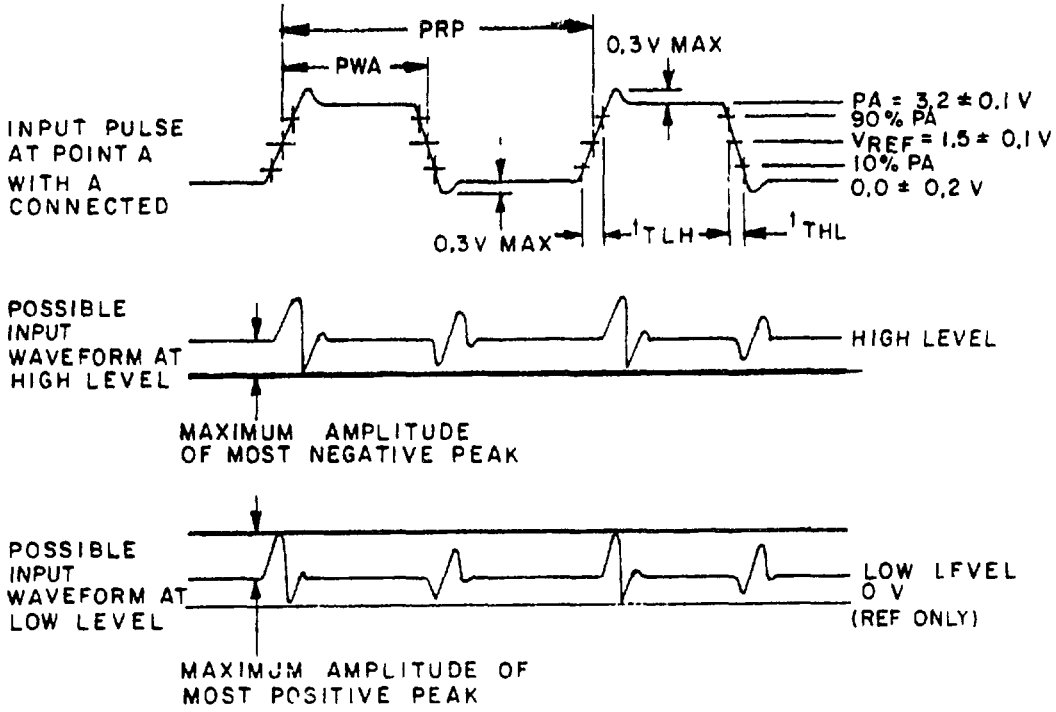
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 150Ω ±5%, 1/4 W.
3. C = Sufficient capacitance to reduce item P01-2 power supply noise to < 5 mV.
4. Load each output with the load as specified on figure 18.
5. For test conditions T1-T8, adjust item P01-2 to 0.65 V ±1%.
6. For test conditions T9-T16, adjust item P01-2 to 2.2 V ±1%.

FIGURE 34. Input crosstalk (low-level and high-level)-test circuit for table XVIII.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 20 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistor R2 and input pins, items YO1-1 and YO1-2 to input pins, load circuits to output pins, and ground leads of items YO1-1 and YO1-2 to test fixture and module ground pins shall be less than 2 inches (50.8 mm).

FIGURE 35. Input crosstalk test (low-level and high-level) waveforms for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	36,37

TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	1.6	3.0	µF	38

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TABLE XXI. Input and output capacitance test.

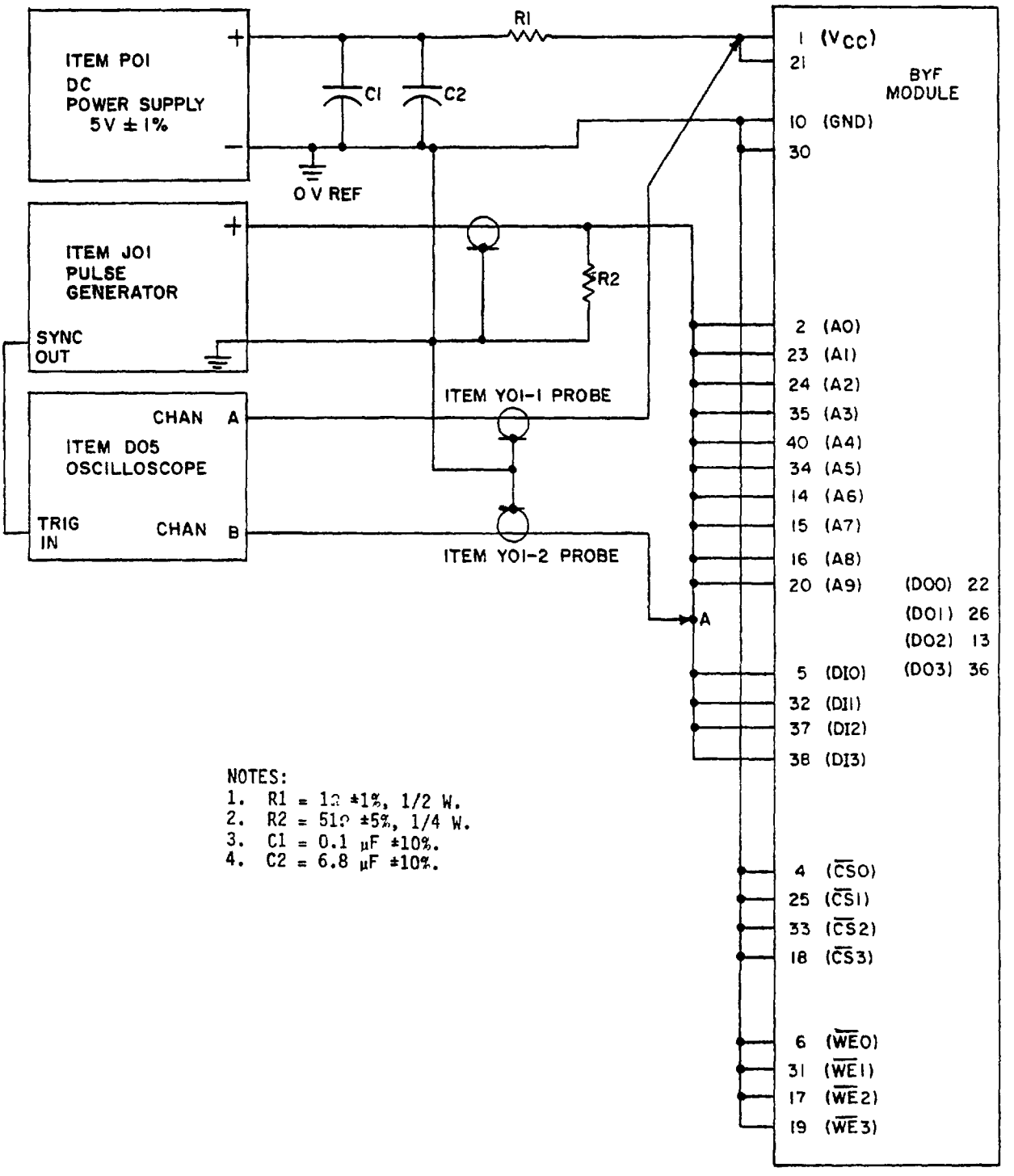
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	$\overline{CS0}$ Input capacitance	4	Open	2	25	pF	38
W2	$\overline{CS1}$ "	25	"	"	"	"	"
W3	$\overline{CS2}$ "	33	"	"	"	"	"
W4	$\overline{CS3}$ "	18	"	"	"	"	"
W5	A0 "	2	"	"	45	"	"
W6	A1 "	23	"	"	"	"	"
W7	A2 "	24	"	"	"	"	"
W8	A3 "	35	"	"	"	"	"
W9	A4 "	40	"	"	"	"	"
W10	A5 "	34	"	"	"	"	"
W11	A6 "	14	"	"	"	"	"
W12	A7 "	15	"	"	"	"	"
W13	A8 "	16	"	"	"	"	"
W14	A9 "	20	"	"	"	"	"
W15	$\overline{WE0}$ "	6	"	"	15	"	"
W16	$\overline{WE1}$ "	31	"	"	"	"	"
W17	$\overline{WE2}$ "	17	"	"	"	"	"
W18	$\overline{WE3}$ "	19	"	"	"	"	"
W19	DI0 "	5	"	"	"	"	"
W20	DI1 "	32	"	"	"	"	"
W21	DI2 "	37	"	"	"	"	"
W22	DI3 "	38	"	"	"	"	"

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TABLE XXI. Input and output capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	DO0 Output capacitance	22	Open	2	25	pF	38
X2	DO1 "	26	"	"	"	"	"
X3	DO2 "	13	"	"	"	"	"
X4	DO3 "	36	"	"	"	"	"

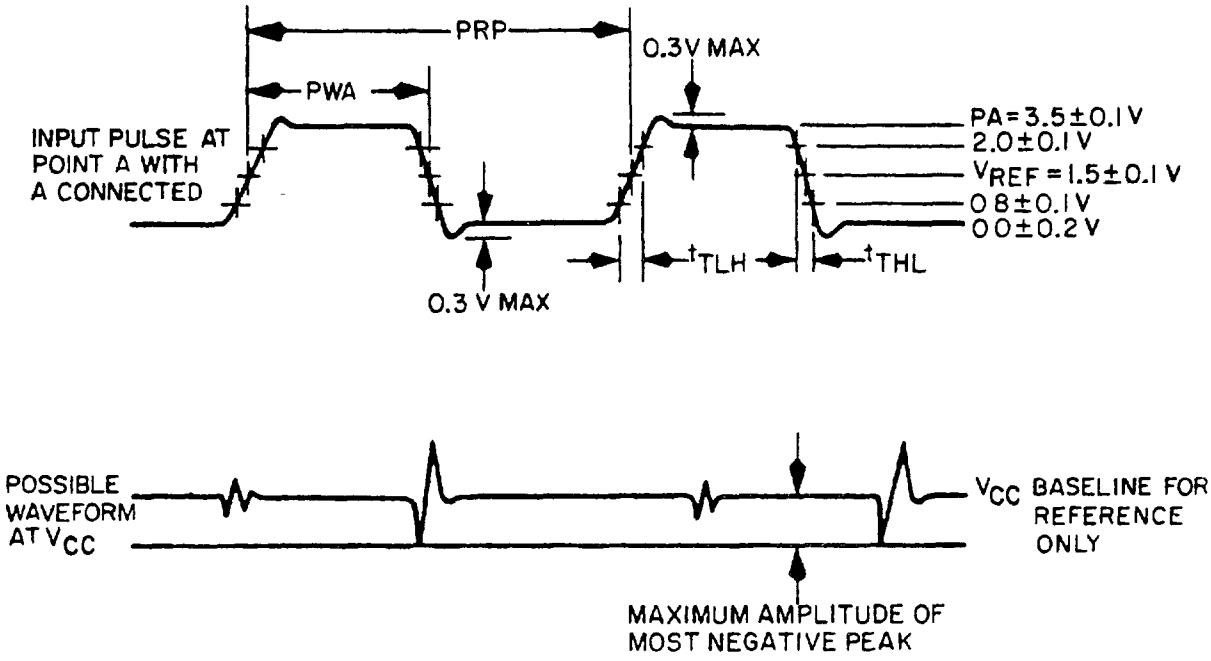
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NOTES:
1. R1 = 1Ω ±1%, 1/2 W.
2. R2 = 51Ω ±5%, 1/4 W.
3. C1 = 0.1 µF ±10%.
4. C2 = 6.8 µF ±10%.

FIGURE 36. Power supply transient amplitude test circuit for table XIX.

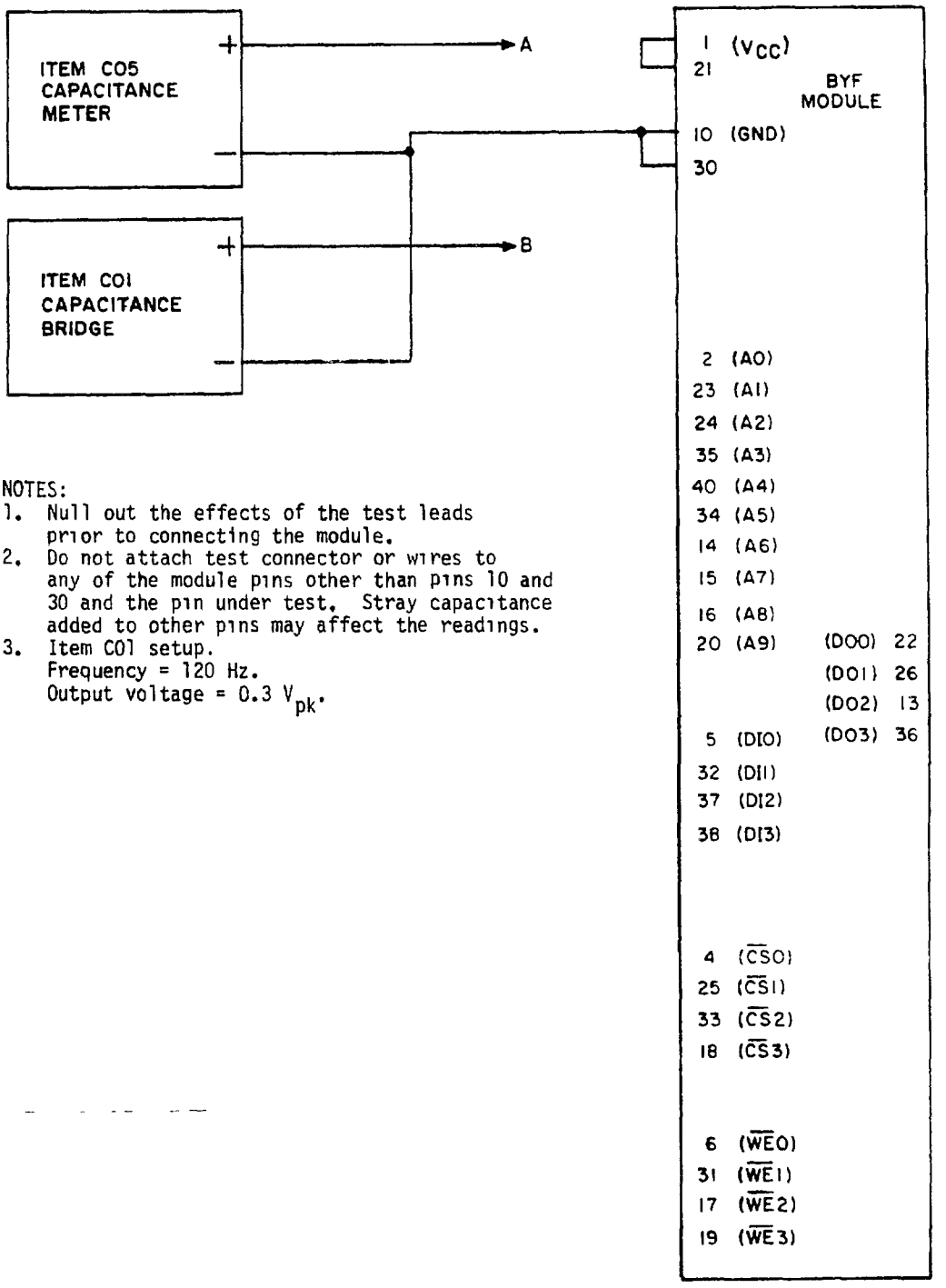
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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns. (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. All voltages are referenced to ground.

FIGURE 37. Power supply transient amplitude test circuit waveforms for table XIX.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10 and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup.
   Frequency = 120 Hz.
   Output voltage = 0.3 V_{pk}.

FIGURE 38. Power supply filter, input, and output capacitance test circuit for tables XX and XXI.

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5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

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